Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

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Part I

Multivariable Calculus

Chapter 1

Differentiation in $\mathbb{R}^n$

1.1 Differentials and Partial Derivatives

Definition 1.1.1: Fréchet Derivative

Let V, W be normed vector spaces. Let U be an open subset of V. Then $f : U \to W$ is called ***Fréchet differentiable*** or just ***differentiable*** at $p \in U$ if there exists a continuous linear map $Df(p) : V \to W$ such that

$$\lim_{h \to 0} \frac{\left\| f(p+h) - f(p) - Df(p)(h) \right\|_W}{\|h\|_V} = 0.$$

We call $Df(p)$ the ***differential of f at p***. We call f ***differentiable*** if it is differentiable at every point $p \in U$. We call f ***continuously differentiable*** if the map $Df : p \mapsto Df(p)$ is continuous.

Recall that linear maps are continuous if and only if they are bounded, which is always the case if V is finite-dimensional.

Exercise 1.1.2. *Verify that if we identify scalars $c \in \mathbb{R}$ with linear maps $x \mapsto c \cdot x$, this definition agrees with the usual definition for $f : \mathbb{R} \to \mathbb{R}$.*

Theorem 1.1.3. If $f : U \to W$ is differentiable, it is continuous.

Proof. If f is differentiable, we have

$$\lim_{\|h\|_V \to 0} \frac{\left\| f(p+h) - f(p) - Df(p)(h) \right\|_W}{\|h\|_V} = 0.$$

This can only be the case if

$$\lim_{\|h\|_V \to 0} \left\| f(p+h) - f(p) - Df(p)(h) \right\|_W = 0,$$

i.e. if

$$\lim_{\|h\|_V \to 0} f(p+h) = \lim_{\|h\|_V \to 0} f(p) + Df(p)(h).$$

Since $Df(p)$ is continous with $Df(0) = 0$, this simplifies to

$$\lim_{\|h\|_V \to 0} f(p+h) = f(p),$$

which is the definition of sequential continuity. □

Definition 1.1.4: Directional and Partial Derivates

Let V and W be normed vector spaces. Let $v \in V$. Let Let $U \subseteq V$ be open and $f : U \to W$. Then we call the limit

$$\frac{\partial}{\partial v} f(p) = \frac{d}{dh} f(p + hv)\bigg|_{h=0} = \lim_{h \to 0} \frac{f(p + hv) - f(p)}{h}$$

the ***directional derivative*** of f at p in direction v. If $(x_0, \ldots, x_n)$ is a basis of V, we call the directional derivatives

$$\partial_i f(p) := \frac{\partial}{\partial x_i} f(p)$$

the ***partial derivatives of f at p***. If all partial derivatives exist, we call f ***partially differentiable***. If all maps $\partial_i f : p \mapsto \partial_i f(p)$ are continuous, we call f ***continuously partially differentiable***.

Partial derivatives aren't actually any more complicated than standard one-dimensional derivatives - you can calculate partial derivatives by simply fixing all coordinates except one and then taking a regular derivative in the remaining coordinate. For example, if we have a function

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R} \\ (x, y, z) &\mapsto f(x, y, z), \end{aligned}$$

we can define a function $f|_1 : \mathbb{R} \to \mathbb{R}$ by restriction to the first coordinate:

$$f|_1(x) := f(x, y, z).$$

Using this function, we can simplify the partial derivative:

$$\begin{aligned} \partial_1 f(x, y, z) &= \lim_{h \to 0} \frac{f((x, y, z) + he_1) - f(x, y, z)}{h} \\ &= \lim_{h \to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h} \\ &= \lim_{h \to 0} \frac{f|_1(x + h) - f(x)}{h} \\ &= \frac{d}{dx} f|_1(x) \end{aligned}$$

This tells us that, for example, the function $f(x, y, z) = x \cdot y \cdot z$ has partial derivatives

$$\begin{aligned} \partial_1 f(x, y, z) &= \frac{d}{dx} xyz = yz \\ \partial_2 f(x, y, z) &= \frac{d}{dy} xyz = xz \\ \partial_3 f(x, y, z) &= \frac{d}{dz} xyz = xy. \end{aligned}$$

Exercise 1.1.5. 1. *Show that if f is differentiable, then all directional derivatives exist and fulfill $\frac{\partial}{\partial v} f(p) = Df(p)(v)$.*

2. *Conclude that $Df(p)$ is represented in basis $(x_0, \dots, x_n)$ by the* **Jacobian matrix**

$$Jf(p) = \left(\frac{\partial}{\partial x_1} f(p), \dots, \frac{\partial}{\partial x_n} f(p) \right) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(p) & \dots & \frac{\partial}{\partial x_n} f_1(p) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(p) & \dots & \frac{\partial}{\partial x_n} f_n(p) \end{pmatrix}$$

If the image of f is one-dimensional, the Jacobian matrix is simply a row vector. We call the transpose of this vector the ***gradient of f at p*** and denote it

$$\nabla f(p) := Jf(p)^\top = \begin{pmatrix} \frac{\partial}{\partial x_1} f(p) \\ \vdots \frac{\partial}{\partial x_n} f(p) \end{pmatrix}$$

The transposition comes from the fact that we would like to be able to get the directional derivatives of f at p by taking a scalar product with the gradient, which gives us

$$\begin{aligned} \left\langle \nabla f(p), v \right\rangle &= \nabla f(p)^\top v \\ &= Jf(p) \cdot v \\ &= Df(p)(v) \\ &= \frac{\partial}{\partial v} f(p) \end{aligned}$$

There is one very important pitfall to watch out for when it comes to differentiability: We have just shown that if f is differentiable, we can get the differential by construction the Jacobian matrix from the partial derivatives. This makes it look like every partially differentiable function is differentiable, which turns out to not be true in general.

For example, there exist functions which are partially differentiable, but not continuous - we showed earlier that all differentiable functions are continuous, so these functions are quick counterexamples.

One example is

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{xy}{(x^2+y^2)^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}. \end{aligned}$$

This function has $\partial_1 f(0) = \partial_2 f(0) = 0$, since $f(h, 0) = f(0, h) = 0$ for all $h \in \mathbb{R}$, but isn't continuous, since $\lim_{n\to\infty} f\left(\frac{1}{n}, \frac{1}{n}\right) = \lim_{n\to\infty} \frac{n^2}{4} = \infty \neq f(0)$.

For a more involved counterexample, consider the function

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function is continuous - for $(x, y) \neq 0$, it is a composition of continuous functions, and thus continuous, and since $x^2 + y^2 \geq y^2$, we additionally have

$$\begin{aligned}\left|f(x,y)\right| &= \frac{\left|y^3\right|}{\left|x^2+y^2\right|} \\ &= \frac{\left|y^3\right|}{x^2+y^2} \\ &\leq \frac{\left|y^3\right|}{y^2} \\ &= \left|y\right|,\end{aligned}$$

so $f(x, y)$ converges to $f(0, 0) = 0$ as (x, y) converges to 0, implying that f is also continuous at the origin.

f is also directionally differentiable in all directions:

$$\begin{aligned}\partial_v f(0) &= \lim_{h\to 0} \frac{f(h\cdot v) - f(0)}{h} \\ &= \lim_{h\to 0} \frac{f(h\cdot v)}{h} \\ &= \lim_{h\to 0} \frac{\left(\frac{h^3 v_2^3}{h^2 v_1^2 + h^2 v_2^2}\right)}{h} \\ &= \lim_{h\to 0} \frac{h^3 v_2^3}{h^3 v_1^2 + h^3 v_2^2} \\ &= \lim_{h\to 0} \frac{v_2^3}{v_1^2 + v_2^2} \\ &= \frac{v_2^3}{v_1^2 + v_2^2} \\ &= f(v)\end{aligned}$$

In particular, $\partial_1 f(0) = 0$ and $\partial_2 f(0) = 1$. However, f is not differentiable at 0, since otherwise our representation $Df(0) = Jf(0)$ would give us:

$$\lim_{\vec{h}\to 0} \frac{\left|f(h) - f(0) - Df(0)(\vec{h})\right|}{\left\|\vec{h}\right\|_2} = \lim_{\vec{h}\to 0} \left|\frac{f(h) - Df(0)(h)}{\left\|\vec{h}\right\|_2}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{f(h) - \partial_1 f(0)\cdot h_1 + \partial_2 f(0)\cdot h_2}{\sqrt{h_1^2 + h_2^2}}\right| = \lim_{\vec{h}\to 0} \left|\frac{f(h) - h_2}{\sqrt{h_1^2 + h_2^2}}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3}{h_1^2 + h_2^2} - h_2\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2(h_1^2 + h_2^2)}{h_1^2 + h_2^2}\right)\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2 h_1^2 - h_2^3}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2 h_1^2}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3}\right|$$

Since we are only interested in whether this expression converges to 0, we can

square it to simplify slightly:

$$\left|\frac{h_2 h_1^2}{\sqrt{h_1^2+h_2^2}^3}\right|^2 = \left|\frac{h_2^2 h_1^4}{(h_1^2+h_2^2)^3}\right|$$

If we now set $h := \left(\frac{1}{n}, \frac{1}{n}\right)$, we get:

$$\left|\frac{h_2^2 h_1^4}{(h_1^2+h_2^2)^3}\right| = \left|\frac{\frac{1}{n^6}}{\left(\frac{2}{n^2}\right)^3}\right| = \left|\frac{\frac{1}{n^6}}{\frac{8}{n^6}}\right| = \frac{1}{8} \neq 0$$

The important takeaway here is that directional differentiability only guarantees "differentiability along straight paths", while full differentiability also requires "differentiability along curved paths".

Theorem 1.1.6

$f : V \to W$ is continuously differentiable if and only if it is continuously partially differentiable.

Proof. $\Rightarrow$: We have $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$. If the differential $Jf : \mathbb{R}^n \mapsto \mathbb{R}^{mn}$ exists, it is continuous if and only if every component $(Jf)_i : \mathbb{R}^n \mapsto \mathbb{R}$ is continuous. By the definition of Jf, these are the partial derivatives.

$\Leftarrow$: Since every normed complex vector space is isomorphic to a normed real vector space, we can assume without loss of generality that W is real. Since all norms are equivalent, and since a sequence in W converges iff. all components converge, we can additionally assume that W is one-dimensional, i.e. $W \cong \mathbb{R}$.

We need to show that

$$\lim_{h \to \vec{0}} \frac{f(p+h) - f(p) - Jf(p)h}{\|h\|_2} = \vec{0}$$

Let $h := \sum_{i=1}^n h_i e_i$. We decompose the path from p to $p+h$ into its partial steps, taking one coordinate direction at a time:

$$\forall k \in [0..n] :$$
$$p_k = p + \sum_{i=1}^{k} h_i e_i$$

Now, by the mean value theorem, there exists an $s_k \in [0,1]$ such that

$$\begin{aligned} f(p_k) - f\left(p_{k-1}\right) &= f(p_{k-1} + h_k e_k) - f\left(p_{k-1}\right) \\ &= \partial_k f(p_{k-1} - s_k h_n e_n) h_k \end{aligned}$$

Now, since

$$Jf(p)(h) = \sum_{i=1}^{n} \partial_i f(p) h_i,$$

we get:

$$\begin{aligned}
&\frac{\left|f(p+h)-f(p)-Jf(p)(h)\right|}{\|h\|_2}\\
=&\frac{1}{\|h\|_2}\left|f(p_n)-f(p_0)-\sum_{i=1}^{n}\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}f(p_i)-f(p_{i-1})-\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\partial_i f(p_{i-1}+s_ih_ie_i)h_i-\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\left(\partial_i f(p_{i-1}+s_ih_ie_i)-\partial_i f(p)\right)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right)h_i\right|\\
\leq&\frac{1}{\|h\|_2}\sum_{i=1}^{n}\left|\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right)h_i\right|\\
=&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right|\cdot\frac{|h_i|}{\|h\|_2}\\
\leq&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right|
\end{aligned}$$

Since $s_i \in [0,1]$, we have

$$\lim_{h\to 0} p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i = p$$

And thus, since the partial derivatives $\partial_i f$ are continuous, the right side converges to zero, so f is differentiable.

□

Exercise 1.1.7. *Let V be an inner product space over a field $\mathbb{F}=\mathbb{R}$ or $\mathbb{F}=\mathbb{C}$. Let $v \in V$. Then the function*

$$\begin{aligned}\langle\cdot, v\rangle : V &\to \mathbb{F}\\ w &\mapsto \langle w, v\rangle\end{aligned}$$

is differentiable, with $D\langle\cdot, v\rangle(w) = v^\top$ for all $w \in V$.

Exercise 1.1.8. *Let b be a symmetric bilinear form. Let*

$$q(v) := b(v,v)$$

be the corresponding quadratic form. Then q is differentiable, with

$$Dq(v)(w) = 2b(v,w)$$

1.2 Rules of Differentiation

Exercise 1.2.1. *Show that the differential operator D is linear, i.e. that we have*

$$D(\alpha f + \beta g)(p) = \alpha D f(p) + \beta D g(p)$$

for f, g differentiable at p.

Theorem 1.2.2: Chain Rule in $\mathbb{R}^n$

Let $f : U \to \mathbb{R}^m$ and $g : V \to \mathbb{R}^r$ with $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$ open and $f(U) \subseteq V$. Let f be differentiable at a point $p \in U$ and let g be differentiable at $f(p)$. Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

Exercise 1.2.3. *Verify that this version of the chain rule agrees with the familiar chain rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. By definition of the differential, we have:

$$\begin{aligned} f(x) &= f(p) \\ &\quad + Df(p)(x-p) \\ &\quad + \varepsilon_f(x)\|x-p\| \\ g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

For some functions $\varepsilon_f : U \to \mathbb{R}^m$, $\varepsilon_g : U \to \mathbb{R}^r$ which are continuous in p and $f(p)$ and fulfill $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$.
The second equation tells us:

$$\begin{aligned} g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

Plugging in our equation for $f(x)$ now lets us regroup our terms into an

equation of the same form as the definition of the differential:

$$
\begin{aligned}
g(f(x)) &= g(f(p)) \\
&\quad + Dg(f(p))\left[\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right] \\
&\quad + \varepsilon_g(f(x))\left\|\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))\left[Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right] \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + Dg(f(p))\varepsilon_f(x)\|x-p\| \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + \underbrace{\left(\varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x))\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right) + \varepsilon_f(x)\right\|\right)}_{:=\varepsilon_{g\circ f}(x)}\|x-p\| \\
&= g(f(p)) \\
&\quad + (Dg(f(p)) \circ Df(p))(x-p) \\
&\quad + \varepsilon_{g\circ f}(x)\|x-p\|
\end{aligned}
$$

All that remains is to show that $\varepsilon_{g\circ f}(x)$ is continuous if we set

$$\varepsilon_{g\circ f}(p) = 0.$$

Since we have $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$, we have our desired result if we can show that

$$\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right)\right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since $\mathbb{R}^n$ is finite-dimensional, we can find a constant c such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof.

□

Theorem 1.2.4: Product Rule in $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$. Let $f, g : U \to \mathbb{R}$ at a point $p \in U$. Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

Exercise 1.2.5. *Verify that this version of the product rule agrees with the familiar product rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. We define the multiplication function $m(x, y) := x \cdot y$. The partial derivatives are $\partial_1 m(x, y) = y$ and $\partial_2 m(x, y) = x$, i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned} D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\ &= Dm((f, g)(p)) \circ D(f, g)(p) \\ &= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\ &= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\ &= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\ &= Df(p) \cdot g(p) + f(p) \cdot Dg(p). \end{aligned}$$

□

Exercise 1.2.6. *Let $f : V \to W$ be differentiable. Let $\gamma : [0,1] \to V$ be differentiable. Then $f \circ \gamma$ is differentiable with*

$$(f \circ \gamma)'(t) = Df(\gamma'(t)) \cdot \gamma(t)$$

Theorem 1.2.7. Let $U \subseteq V$ be open. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is locally constant.

Corollary 1.2.8. Let $U \subseteq V$ be open and connected. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is constant.

1.3 Higher Partial Derivatives

Theorem 1.3.1: Schwartz's Theorem

Let $f : U \subseteq V \to W$ be twice continuously partially differentiable. Then the order of partial derivation does not matter, i.e.

$$\partial_i \partial_j f = \partial_j \partial_i f$$

for all $i, j \in [1..n]$.

Theorem 1.3.2: First-Order Taylor Expansion in multiple Variables

Let $f : U \subseteq V \to W$. Let f be continuously differentiable. Let $x \in U$, $h \in V$ fulfill $[x, x + h] \subseteq V$. Then

$$f(x + h) = f(x) + Df(x)(h) + r(h),$$

with $r \in o(\|h\|)$.

1.4 Differential Operators

Chapter 2

The Two Queens of Multivariable Calculus

2.1 The Inverse Function Theorem

Theorem 2.1.1: Inverse Function Theorem

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $x \in U$ be a point such that $Df(x) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible. Then there exists an open neighbourhood V of x such that:

1. $f(V)$ is an open neighbourhood of $f(x)$.
2. $f|_V$ is a C^1-diffeomorphism, i.e. invertible with continuously differentiable inverse.

In short: if a continuously differentiable function has an invertible differential at a given point, there exists an open neighbourhood around that point in which the function is invertible:

Corollary 2.1.2. Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $Df(x)$ be invertible for all $x \in \mathbb{R}^n$. Then $f(U)$ is open.

2.2 The Implicit Function Theorem

Theorem 2.2.1: Implicit Function Theorem

Let $U \subseteq \mathbb{R}^m \times \mathbb{R}^k$ be open. Let $f : U \to \mathbb{R}^k$ be continuously differentiable. Let $(x, y) \in U$ with $f(x, y) = z$. Let $Df|_{\mathbb{R}^k}(x, y) : \mathbb{R}^k \to \mathbb{R}^k$ be invertible. Then there exist

1. open neighbourhoods V of x and W of y,
2. and a continuously differentiable function $g : V \to W$,

such that the set of values in $V \times W$ for which f takes the value z is the graph of g, i.e.

$$\{(v, w) \in V \times W \mid f(v, w) = z\} = \{(v, g(v)) \mid v \in V\}$$

In short: under mild assumptions, the level sets of a continuously differentiable function are graphs of continuously differentiable functions:

Chapter 3

Multiple Integrals

Lemma 3.0.1. Let $K \subseteq \mathbb{R}^n$ be compact. Let $f : K \times [a,b] \to \mathbb{R}$ be continuous. Then the function

$$x \mapsto \int_a^b f(x,t)\,dt$$

is uniformly continuous.

Proof. $K \times [a,b]$ is a finite product of compact sets, i.e. compact by Theorem **??**. Thus, f is uniformly continuous by Theorem **??**, which means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any $x, y \in \mathbb{R}, t \in \mathbb{R}^n$

$$d((x,t),(y,t)) < \delta \implies d(f(x,t), f(y,t)) < \varepsilon.$$

This tells us that for any $x, y \in \mathbb{R}$,

$$\begin{aligned} d(x,y) < \delta &\implies d((x,t),(y,t)) < \varepsilon \\ &\implies \left|f(x,t) - f(y,t)\right| = d(f(x,t), f(y,t)) < \varepsilon \\ &\implies d\left(\int_a^b f(x,t)\,dt, \int_a^b f(y,t)\,dt\right) \\ &\quad = \left|\int_a^b f(x,t)\,dt - \int_a^b f(y,t)\,dt\right| \\ &\quad \le \int_a^b \left|f(x,t) - f(y,t)\right| dt \\ &\quad \le (b-a)\varepsilon, \end{aligned}$$

since ε was arbitrary, our function is uniformly continuous as desired. □

Definition 3.0.2. Given a compact cuboid $Q = \prod_{i=1}^n [a_i, b_i] \subseteq \mathbb{R}^n$ and a continuous function $f : Q \to \mathbb{R}$, we declare the ***integral of f over Q*** to be the nested integral

$$\int_Q f(x)\,d^n x := \int_{a_n}^{b_n} \dots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

By Lemma 3.0.1, all involved functions are continuous, and thus their integrals

are indeed well-defined. This definition will become a fully justified theorem once we begin our study of the Lebesgue integral later on.

Exercise 3.0.3. *Verify that linearity and monotonicity carry over from the single-variable case, i.e show that for any $f, g : Q \to \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have:*

1. $\int_Q f(x) + g(x)\, d^n x = \int_Q f(x)\, d^n x + \int_Q g(x)\, d^n x$
2. $\int_Q \lambda f(x)\, d^n x = \lambda \int_Q f(x)\, d^n x$
3. $f \leq g \implies \int_Q f(x)\, d^n x \leq \int_Q g(x)\, d^n x$
4. $\left|\int_Q f(x)\, d^n x\right| \leq \int_Q \left|f(x)\right| d^n x$

Definition 3.0.4. We define

$$\text{vol } Q = \int_Q 1 = \prod_{i=1}^{n} |b_i - a_i|$$

Definition 3.0.5. By partitioning the intervals $[a, b] \times [c, d] := Q \subseteq \mathbb{R}^2$ into r equidistant chunks with boundaries

$$a = a_1 \leq a_2 \leq \ldots \leq a_r = b$$
$$c = c_1 \leq c_2 \leq \ldots \leq c_r = d,$$

we partition Q into r^2 equally sized smaller rectangles

$$Q_{ij} := [a_i, c_{i+1}] \times [a_j, c_{j+1}]$$

which must each have area $\text{vol } Q/r^2$. We can thus define ***two-dimensional Riemann sums*** analogously to how we did it in one dimension:

$$S_r^2(f) := \sum_{i=0}^{r-1} \sum_{j=0}^{r-1} f(a_i, c_j) \frac{\text{vol } Q}{r^2}$$

Lemma 3.0.6: Multivariable Integration via Riemann Sums

Given a compact cuboid $Q \subseteq \mathbb{R}^2$ and a continuous function $f : Q \to \mathbb{R}$, we have

$$\int_Q f(x)\, d^2 x = \lim_{r \to \infty} S_r^2(f)$$

Proof. Applying the single-variable version of this theorem twice, we get:

$$\begin{aligned}
\int_Q f(x)\, d^2 x &= \int_c^d \int_a^b f(x_1, x_2)\, dx_1\, dx_2 \\
&= \lim_{r \to \infty} S_r \left(\int_a^b f(x_1, x_2)\, dx_1 \right) \\
&= \lim_{r \to \infty} S_r^2 \big(f(x_1, x_2) \big)
\end{aligned}$$

□

Part II

Point-Set Topology

Chapter 4

Topological Spaces

4.1 Topologies

Definition 4.1.1: Accumulation points and isolated points

Let X be a topological space. Let $A \subseteq X$. Then we call $x \in X$ an ***accumulation point*** of A if every neighbourhood of x contains a point in A ***other than x itself***.
If a point $x \in A$ isnt an accumulation point of A, we call x an ***isolated point***.

Definition 4.1.2. Let X be a topological space. Let $A \subseteq X$. Then the set of all accumulation points of A is denoted A', and is known as the ***derived set of A***.

Exercise 4.1.3. *Show that* $A' \subseteq \overline{A}$.

4.2 Basic Separation Properties

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic",
Steven Vickers

Definition 4.2.1. Let $(X, \mathcal{O})$ be a topological space. Let $x, y \in X$. Then we call x and y ***topologically distinguishable*** iff. $\mathcal{U}(x) \neq \mathcal{U}(y)$, i.e. one of the points has a neighborhood not containing the other.

Definition 4.2.2: $T0$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***To*** if every pair of points is topologically distinguishable.

Definition 4.2.3: $T1$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T1*** if for every pair of points, both of them have neighbourhoods not containing the other:

$$\begin{aligned} &\forall x, y \in X : \\ &\exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ &\quad x \notin U_y, y \notin U_x. \end{aligned}$$

Exercise 4.2.4.
1. *Find a topological space that isn't To.*
2. *Show that every T1 space is To.*
3. *Find a space that is To, but not T1.*

Exercise 4.2.5. *Show that $(X, \mathcal{O})$ is T1 if and only if every singleton set is closed.*

Exercise 4.2.6. *Show that $(X, \mathcal{O})$ is T1 if and only if every subset of X is the intersection of all of its neighborhoods:*

$$\forall Y \subseteq X : Y = \bigcap \{U \mid U \in \mathcal{U}(Y)\}$$

Definition 4.2.7: Hausdorff Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T2*** or ***Hausdorff*** if every pair of points can be housed orf from each other, i.e. they have disjoint neighbourhoods:

$$\begin{aligned} &\forall x \neq y \in X : \\ &\exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ &\quad U_x \cap U_Y = \emptyset \end{aligned}$$

Exercise 4.2.8. *Show that every Hausdorff space is T1.*

Exercise 4.2.9. *Let $(X, \mathcal{O})$ be Hausdorff. Then every convergent sequence in X has a unique limit.*

Exercise 4.2.10. *Let X be a Hausdorff space. Let x be an accumulation point of $A \subseteq X$. Then every neighbourhood of x contains infinitely many points of A.*

Lemma 4.2.11. Let X be a Hausdorff space. Then the set of accumulation points of any subset is closed.

Proof. Let $A \subseteq X$. Let A' be the set of accumulation points of A. Let x be an accumulation point of A'. Since X is Hausdorff, every neighbourhood of x must contain infinitely many points $x_i \in A'$. But since these neighbourhoods are also neighbourhoods of the x_i, they must also all contain infinitely many points in A.

Therefore, x is already an accumulation point of A, i.e. $x \in A'$. Therefore, A' contains all of its accumulation points, and is thus closed. □

Exercise 4.2.12. *Find a space X and a subset A such that the set of accumulation points of A isn't closed.*

Theorem 4.2.13. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every point is the intersection of the closures of all of its neighbourhoods:

$$\forall x \in X : \{x\} = \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}$$

Proof. $\Rightarrow$ Let $(X, \mathcal{O})$ be Hausdorff. Let $x, y \in X$ and $y \notin \{x\}$, i.e. $y \neq x$. Then, by Hausdorff, there exist disjoint neighborhoods U_x and U_y of x and y. Since U_x and U_y are disjoint, we have $U_x \subseteq X \setminus U_y$. But then, by monotonicity of the closure operator, we get

$$\overline{U}_x \subseteq \overline{X \setminus U_y} = X \setminus U_y,$$

i.e. $y \notin \overline{U}_x$, i.e.

$$y \notin \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}.$$

$\Leftarrow$ Let $x \neq y$. Let $\{x\} = \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}$. Then there must be an open neighborhood U_x of x whose closure doesn't contain y. But then the complement $U_y := X \setminus \overline{U}_x$ of that closure is an open neighborhood of y that is disjoint from U_x.

□

4.3 The Subspace Topology

Theorem 4.3.1

Let $(X, \mathcal{O})$ be a topological space, and let $S \subseteq X$ be any subset. Then the collection

$$\mathcal{O}_S := \{U \subseteq S \mid \exists V \in \mathcal{O} : U = S \cap V\},$$

consisting of all intersections of open sets of X with S, forms a topology, which we call the ***subspace topology***.

Exercise 4.3.2. *Let $S \subseteq X$. Show that $B \subseteq S$ is closed in S if and only if it is equal to the intersection of S with some closed subset of X.*

Exercise 4.3.3. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *If U is open in X, then U is open in S.*
2. *If U is closed in X, then U is closed in S.*

Exercise 4.3.4. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *If U is open in S, and S is open in X, then U is open in X.*
2. *If U is closed in S, and S is closed in X, then U is closed in X.*

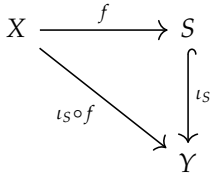
If $U \subseteq S \subseteq X$, the notations $\overline{U}$, $\operatorname{int} U$, and $\operatorname{ext} U$ will always refer to the closure, interior, and exterior of U in X, not in S.

Exercise 4.3.5. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *The closure of U in S is equal to $\overline{U} \cap S$.*
2. *Find an example where the interior of U in S is not equal to $\operatorname{int} U \cap S$.*
3. *Show that $\operatorname{int} U \cap S$ is a subset of the interior of U in S.*

Theorem 4.3.6: Universal Property of the Subspace Topology

Let X and Y be topological spaces. Let $S \subseteq Y$. Then $f : X \to S$ is continuous if and only if the embedded map $\iota_S \circ f : X \to Y$ is continuous.



Proof. $\Rightarrow$: Suppose that $f : X \to S$ is continuous. Let $V \subseteq Y$ be open. Then we have

$$\begin{aligned}(\iota_S \circ f)^{-1}(V) &= f^{-1}(\iota_S^{-1}(V)) \\ &= f^{-1}(S \cap V) \qquad (??)\end{aligned}$$

which is open in X since $S \cap V$ is open in S.

$\Leftarrow$:

□

Corollary 4.3.7. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is continuous.

Corollary 4.3.8. Let X and Y be topological spaces. Let $f : X \to Y$ be continuous. Then:

1. If $S \subseteq X$, the restriction
$$f|_S : S \to Y$$
is continuous.
2. If $f(X) \subseteq T \subseteq Y$, the map
$$f : X \to T$$
is well-defined and continuous.
3. If $Y \subseteq Z$, then the embedded function
$$\iota_Y \circ f : X \to Z$$
is continuous.

Exercise 4.3.9. *Let X be a topological space. Let $R \subseteq S \subseteq X$. Then the subspace topologies of R in S and X agree.*

Exercise 4.3.10.

1. *Every subspace of a To space is To.*
2. *Every subspace of a T1 space if T1.*
3. *Every subspace of a Hausdorff space is Hausdorff.*

Definition 4.3.11

A map that is a homeomorphism onto its image is called a ***topological embedding***, or just ***embedding***.

Exercise 4.3.12. *A homeomorphism is a surjective topological embedding.*

Exercise 4.3.13. *Let X be a topological space. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is an embedding.*

Exercise 4.3.14. *Every embedding is continuous and injective.*

Exercise 4.3.15. *Every injective continuous map that is open or closed is an embedding.*

Exercise 4.3.16. *Find an embedding that is neither open nor closed.*

Theorem 4.3.17

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is homeomorphic to U.

Proof. Let $\Phi_f : U \to \mathbb{R}^{n+k}$ be the canonical map

$$\Phi_f(x) = (x, f(x))$$

taking U to its graph.
By definition of the graph $\Gamma(f)$, $\Phi_f(x) : U \to \Gamma(f)$ is surjective.
Since Φ_f is a composition of continuous maps, it is continuous.
Since the first n components of Φ_f are the identity, Φ_f is injective.
Finally, the projection $p : \mathbb{R}^{n+k} \to \mathbb{R}^n$ is continuous, so the restriction $p|_{\Gamma(f)}$ is a continuous inverse of Φ_f. □

4.4 Bases and Countability

Exercise 4.4.1. *Let $(X, \mathcal{O})$ be a topological space. Let $S \subseteq X$. Then if $\mathcal{B}$ is a basis of $\mathcal{O}$, the collection*

$$\mathcal{B}_S = \{B \cap S \mid B \in \mathcal{B}\}$$

is a basis of the subspace topology $\mathcal{O}_S$.

Definition 4.4.2

We call a topological space ***first countable*** if it has a countable neighbourhood basis around every point x, i.e. a set of neigbourhoods B_i of x such that every neighbourhood of x contains some B_i.

Exercise 4.4.3. *Every metric space is first countable.*

It turns out that first countable spaces are exactly the spaces where sequences are sufficient to maintain our intuitions about how sequence convergence interacts with topological properties:

Theorem 4.4.4: Detecting openness and closedness using sequences

Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. $x \in \operatorname{int} A$ if and only if every sequence in X converging to x is eventually in A.
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x.

Proof. Copy the proof of the metric case, replacing open balls with our countable neighbourhood basis. □

Corollary 4.4.5. Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A.
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A.

Definition 4.4.6

We call a topological space ***separable*** if it contains a countable dense subset.

Exercise 4.4.7.

1. *Every countable space is separable.*
2. *$\mathbb{R}^n$ is separable for every $n \in \mathbb{N}$.*

Definition 4.4.8

We call a topological space ***second countable*** if it has a countable basis.

Exercise 4.4.9. *$\mathbb{R}^n$ is second countable for every $n \in \mathbb{N}$.*

Exercise 4.4.10. *Let X be second countable. Then:*

1. *X is first countable.*
2. *X is separable.*

Theorem 4.4.11. A metric space is second countable if and only if it is separable.

Exercise 4.4.12. *Let $H = \left\{(0, y) \,\middle|\, y \in \mathbb{R}_{\geq 0}\right\} \subseteq \mathbb{R}^2$ be the closed upper half plane. We define a topology $\mathcal{O}$ on H by taking as a basis the open subsets of $\mathbb{R}^2$ that are contained in H and intersect the boundary of H in at most one point. Show that $(H, \mathcal{O})$ is separable, but not second countable.*

Chapter 5

Compactness

The familiar notions of open covers and compactness generalize directly from metric spaces to topological spaces. The most major difference is that sequential compactness is generally weaker than "covering compactness", and has to be replaced with a notion of compactness based on ultrafilter convergence.

Definition 5.0.1. Let $(X, \mathcal{O})$ be a topological space. Then we call a set $\mathcal{U} \subseteq \mathcal{O}$ of open subsets an ***open covering*** of X if every point of X is contained in the union of all elements of $\mathcal{U}$, i.e. if

$$\bigcup_{U_i \in \mathcal{U}} U_i = X$$

Definition 5.0.2: Compactness

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***compact*** if every open covering of X contains a finite open covering as a subset. We call a subset $K \subseteq X$ compact if it is compact as a topological space equipped with the subspace topology.

Exercise 5.0.3. *Every closed subset of a compact space is compact.*

Exercise 5.0.4. *Let $(X, \mathcal{O})$ be compact and $f : X \to Y$ be continuous. Then $f[X]$ is compact.*

Exercise 5.0.5. *Let $(X, \mathcal{O})$ be a nonempty topological space. Then the following are equivalent:*

1. *$(X, \mathcal{O})$ is compact, i.e. every open covering has a finite subcovering.*
2. *Every set $\mathcal{A}$ of closed sets whose intersection is the empty set has a finite subset whose intersection is the empty set.*
3. *If $\mathcal{A}$ is a set of closed sets such that the intersection over any finite subset of $\mathcal{A}$ is non-empty, the intersection of $\mathcal{A}$ is non-empty.*
4. *Every filter on $(X, \mathcal{O})$ has a limit point.*
5. *Every ultrafilter on $(X, \mathcal{O})$ is convergent.*

Theorem 5.0.6. ***(Alexander Subbase Theorem)*** Let $(X, \mathcal{O})$ be a topological space. Let $\mathcal{S}$ be a subbasis. Then X is compact if every open covering of X by sets of $\mathcal{S}$ has a finite subcovering.

Proof. Assume that X is not compact. Then there exists a non-convergent ultrafilter $\mathcal{F}$, i.e. we have

$$\begin{aligned}&\forall x \in X : \\ &\exists U_x \in \mathcal{U}(x) : \\ &\quad (X \setminus U_x) \in \mathcal{F}\end{aligned}$$

By definition of subbases, every $U_x \in \mathcal{U}(x)$ contains an open subset O_x of the form

$$O_x = \bigcap_{k=0}^{n} S_{k,x}$$

for some $S_{k,x} \in \mathcal{S}$ and $n \in \mathbb{N}$. We therefore have

$$\begin{aligned}X \setminus U_x &\subseteq X \setminus O_x \\ &= X \setminus \bigcap_{k=0}^{n} S_{k,x}\end{aligned}$$

which implies $X \setminus \bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ by the filter axioms. Since the intersection of this set with $\bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ is empty, and filters are closed under finite intersection, $\mathcal{F}$ cannot contain every $S_{k,x}$. Therefore, since $\mathcal{F}$ is an ultrafilter, it must contain at least one set $X \setminus S_{k,x}$ for every x. We will refer to these $S_{k,x}$ with $X \setminus S_{k,x} \in X$ as S_x.

Since every S_x is an open neighborhood of x, and thus contains x, the set of all S_x is an open convering of X. By compactness, there must be a finite subcovering

$$\bigcup_{x \in X'} S_x = X.$$

Now, since $\mathcal{F}$ contains every $X \setminus S_x$, it also contains

$$\bigcap_{x \in X'} (X \setminus S_x) = X \setminus \bigcup_{x \in X'} S_x = X \setminus X = \emptyset.$$

which contradicts $\mathcal{F}$ being a filter. □

Lemma 5.0.7. Let $(X, \mathcal{O})$ be a Hausdorff space. Let $K \subseteq X$ be compact. Then K can be separated from any point $x \in X \setminus K$ by open neighborhoods, i.e:

$$\begin{aligned}&\forall x \in X : \\ &\exists U_x \in \mathcal{U}(x), U_K \in \mathcal{U}(K) : \\ &\quad U_x \cap U_K = \emptyset\end{aligned}$$

Exercise 5.0.8. *Let $(X, \mathcal{O})$ be compact and Hausdorff. Let $A, B \subseteq X$ be closed. Then A and B can be separated from each other by disjoint open neigborhoods.*

Exercise 5.0.9. *Every compact subset of a Hausdorff space is closed.*

Exercise 5.0.10. *Every continuous function from a compact space into a Hausdorff space is closed.*

Exercise 5.0.11. *Every continuous injective function from a compact space into a Hausdorff space is an embedding.*

Definition 5.0.12. We call a topological space X ***Lindelöf*** if every open cover of X has a countable subcover.

Theorem 5.0.13. Every second-countable space is Lindelöf.

Definition 5.0.14. We call a topological space X ***countably compact*** if every countable open cover of X has a finite subcover.

Exercise 5.0.15. *A countably compact space is compact if and only if it is Lindelöf.*

Theorem 5.0.16. A metric space is Lindelöf if and only if it is second countable.

Theorem 5.0.17. Every sequentially compact space is countably compact.

Theorem 5.0.18. A first-countable space is countably compact if and only if it is sequentially compact.

Theorem 5.0.19. In a second countable Hausdorff space, compactness, countable compactness and sequential compactness are all equivalent to each other.

Chapter 6

New Spaces from Old

6.1 Initial Topologies and Product Spaces

Lemma 6.1.1. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if the diagonal

$$\Delta := \{(x, x) \mid x \in X\}$$

is closed in the product topology.

Proof. $\Rightarrow$ Assume $(X, \mathcal{O})$ is Hausdorff. Let $(x, y) \notin \Delta$. Then $x \neq y$ by definition of Δ. Therefore, there exist disjoint neighbourhoods U_x, U_y of x and y. Since these neighborhoods are disjoint, we have

$$\Delta \cap (U_x \times U_y) = \emptyset$$

. Therefore, every point $(x, y) \notin \Delta$ has an open neighborhood $(U_x \times U_y)$ that doesn't intersect Δ, i.e. Δ is closed.

$\Leftarrow$ Let $x \neq y$. Assume that Δ is closed. Then $(x, y) \notin \Delta$ has a basic open neighborhood $U_x \times U_y$ that doesn't intersect Δ. Since $U_x \times U_y$ doesn't intersect Δ, U_x and U_y must be disjoint.

□

6.2 Final Topologies and Quotient Spaces

Definition 6.2.1: Final Topologies

Let X be a topological space. Let Y be any set. Let $\mathcal{F} = \left\{ f_i : X \to Y \;\middle|\; i \in I \right\}$ be a collection of functions. Then the ***final topology on Y with respect to $\mathcal{F}$*** is the finest topology such that all functions are continuous, i.e. $U \subseteq Y$ is open in Y if and only if $f_i^{-1}[U]$ is open in X for all $i \in I$.

A final topology with respect to a single surjective function $q : X \to Y$ is known as a ***quotient topology***, and a space Y equipped with a quotient topology is known as a ***quotient space***.

The name "quotient space" comes from the fact that if $q : X \to Y$ is surjective, then each $y \in Y$ can be viewed as an equivalence class $y = \left\{ x \in X \;\middle|\; q(x) = y \right\}$ of the equivalence relation $x_1 \sim x_2 \leftrightarrow q(x_1) = q(x_2)$, i.e. every surjective functions $q : X \to Y$ can be viewed as the projection function of some equivalence relation. Vice versa, every projection function is surjective by definition.

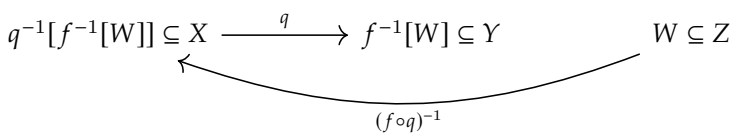
Definition 6.2.2. A continuous surjection $q : X \to Y$ between topological spaces is called a ***quotient map*** or an ***identification map*** if the topology on Y corresponds to the quotient topology induced by q.

Note that this means that we need that V is open in Y if and only if $q^{-1}(V)$ is open in X. We already get one direction for free - since q is continuous, V being open in Y implies that $q^{-1}(V)$ is open in X. Therefore, a continuous surjection is a quotient map if and only if openness of $q^{-1}(V)$ implies openness of V.

Theorem 6.2.3. A surjection $q : X \to Y$ is a quotient map if and only if for all functions $f : Y \to Z$, continuity of $f \circ q : X \to Z$ implies continuity of g and vice versa.

Proof. $\Rightarrow$: Let $q : X \to Y$ be a quotient map. Then q is continuous by definition. Thus, if f is also continuous, then the composition $f \circ q$ must also be continuous.

Vice versa, let $f \circ q$ be continuous. Then, starting with an open set $W \subseteq Z$, $q^{-1}[f^{-1}[W]] = (f \circ q)^{-1}$ is open in Y, since $f \circ q$ is continuous. Since q is a quotient map, a set $q^{-1}[U]$ is open in Y if and only if U is open in X, so $q^{-1}[f^{-1}[W]]$ can only be open in Y if $f^{-1}[W]$ was already open in Y. Thus, f is continuous.



$\Leftarrow$: Let $(Y, \mathcal{O}_Y)$ be the topological space Y equipped with whatever topology it comes packed with.

First of all, note that $q : X \to (Y, \mathcal{O}_Y)$ must be continuous, since we

can simply take $f = 1_Y : (Y, \mathcal{O}_Y) \to (Y, \mathcal{O}_Y)$, so that continuity of 1_Y implies continuity of $q = 1_Y \circ q$.

Now, let $(Y, \mathcal{O}_q)$ be the topological space we get by equipping the set Y with the quotient topology induced by q.

Let $f = 1_Y : (Y, \mathcal{O}_Y) \to (Y, \mathcal{O}_q)$ be the identity function between our two different realizations of Y as a topological space. Then the composition $f \circ q : X \to (Y, \mathcal{O}_q)$ is just $q : X \to (Y, \mathcal{O}_q)$, which is continuous by definition, since $\mathcal{O}_q$ is the quotient topology induced by q. Thus, f must also be continuous.

Finally, the function f^{-1} is continuous, since $f^{-1} \circ q$ is just $q : X \to (Y, \mathcal{O}_Y)$. Therefore, $f : (Y, \mathcal{O}_Y) \cong (Y, \mathcal{O}_q)$ is a homeomorphism, i.e. $\mathcal{O}_Y$ already carries the quotient topology induced by q, i.e. q is a quotient map.

□

Definition 6.2.4: Collapsing a Subspace

Let X be a topological space. Let $A \subseteq X$. Then X/A denotes the quotient space obtained by "collapsing A into a single point", i.e. by the equivalence relation whose equivalence classes are A and all of the points $\{x\} \in X \setminus A$.

Theorem 6.2.5. Let X be regular. Let $A \subseteq X$ be closed. Then $X \setminus A$ is Hausdorff.

Theorem 6.2.6. Let X be normal. Let $A \subseteq X$ be closed. Then $X \setminus A$ is normal.

Theorem 6.2.7. The result of taking the mantle $S^n \times [0,1]$ of an $n+1$-dimensional cylinder and collapsing its bottom is homeomorphic to an $n+1$-dimensional closed ball:

$$S^n \times [0,1]/S^n \times \{0\} \cong \overline{B^{n+1}}$$

Proof. Take the continuous function

$$\begin{aligned} f : S^n \times [0,1] &\to \overline{B^{n+1}} \\ (x,t) &\mapsto tx \end{aligned}$$

We have $f(S^n \times \{0\}) = 0$. Therefore, we can factor our map through our equivalence relation to get a well-defined bijection

$$\begin{aligned} \overline{f} : S^n \times [0,1]/S^n \times \{0\} &\to \overline{B^{n+1}} \\ [x,t] &\mapsto tx. \end{aligned}$$

We have $\overline{f} = f \circ q$, so $\overline{f}$ is a composition of continuous functions, i.e. continuous. Furthermore, $\overline{B^{n+1}}$ is a subspace of $\mathbb{R}^{n+1}$, i.e. Hausdorff, and $S^n \times [0,1]$ is a product of compact sets, i.e. compact. Thus, $\overline{f}$ is a continuous bijection from a compact space into a Hausdorff space, i.e. a Homeomorphism. □

Theorem 6.2.8. Taking a closed n-dimensional ball $\overline{B^n}$ and collapsing its boundary gives an n-sphere:

$$\overline{B^n}/S^{n-1} \cong S^n$$

Definition 6.2.9: Saturated Sets

Let $q : X \to Y$ be a map between topological spaces. Then we call a subset $U \subseteq X$ ***saturated with respect to q*** if

$$\exists V \subseteq Y : U = q^{-1}(V).$$

By extension, given an equivalence relation $\sim$ on X, we call a subset $U \subseteq X$ ***saturated with respect to*** $\sim$ if it is saturated with respect to the induced quotient map, i.e. if

$$U = q^{-1}(V) = \bigcup_{u \in U} [u]_\sim.$$

We call the set $\bigcup_{u\in U}[u]_\sim$ the ***saturation of*** U.

Theorem 6.2.10. A continuous surjection $q : X \to Y$ is a quotient map if and only if it takes saturated open subsets to open subsets or saturated closed subsets to closed subsets.

Proof. 1. Assume that $q : X \to Y$ is a quotient map, i.e. $V \subseteq Y$ is open if and only if $q^{-1}(V) \subseteq X$ is open.

Then if $U \subseteq X$ is saturated, we get $q^{-1}(q(U)) = U$. This immediately tells us that images of saturated open sets are open. If $C \subseteq X$ is closed and saturated, then its complement must be open and saturated, so surjectivity gives us

$$\begin{aligned} q^{-1}(Y \setminus q(C)) &= q^{-1}(q(X) \setminus q(C)) \\ &= q^{-1}(q(X \setminus C)) \\ &= X \setminus C, \end{aligned}$$

i.e. $Y \setminus q(C)$ is open, i.e. $q(C)$ is closed.

2. We have just shown that a continuous surjection taking saturated open sets to open sets also takes saturated closed sets to closed sets. Therefore, assume without loss of generality that q is a continuous surjection taking saturated open sets to open sets.

   If $V \subseteq Y$ is open, continuity tells us directly that $q^{-1}(V) \subseteq X$ is open. On the other hand, given an arbitrary subset $V \subseteq Y$, $q^{-1}(V)$ is saturated by definition, so if it is additionally open, then $q(q^{-1}(V))$ must be open as well. Since q is surjective, we have $q(q^{-1}(V)) = V$, so V is open.

□

Theorem 6.2.11. Let $q : X \to Y$ be a quotient map. Let $U \subseteq X$ be saturated with respect to q and open or closed. Then $q|_U : U \to q(U)$ is a quotient map.

Proof. Since U is saturated and open, we can write $U = q^{-1}(V)$ for some open subset $V \subseteq Y$. The problem then becomes showing that

$$p := q|_{q^{-1}(V)} : q^{-1}(V) \to V$$

is a quotient map, i.e. that p takes saturated open sets to open sets.
Thus, let

$$p^{-1}(W) = q^{-1}(W) \cap q^{-1}(V) \subseteq q^{-1}(V)$$

be an arbitrary open subset of $q^{-1}(V)$ that's saturated with respect to p.
Since preimages play nicely with intersection, we have

$$p^{-1}(W) = q^{-1}(W) \cap q^{-1}(V) = q^{-1}(V \cap W),$$

so $p^{-1}(W)$ is also saturated with respect to q.
Since $q^{-1}(V)$ is open in X, its open subsets are open in X as well, including $q^{-1}(W \cap V)$. Since q is a quotient map, this implies that $W \cap V$ is open in Y, and since $q^{-1}(W \cap V) \subseteq q^{-1}(V)$, we get $W \cap V \subseteq V$, so $W \cap V$ is open in V as well. □

Definition 6.2.12. Let $A \subseteq X$ be topological spaces. Then a continuous map $f : X \to A$ such that $f(a) = a$ for all points $a \in A$ is called a ***retraction***. If such a map exists, A is called a ***retract of X***.

6.3 Adjunction Spaces

Definition 6.3.1: Attaching Spaces along a Subset

Let X and Y be topological spaces, let $A \subseteq X$ be closed, and $f : A \to Y$ be continuous. Then we denote by

$$Y \cup_f X = (X \amalg Y)/\sim$$

the quotient of the disjoint union $X \amalg Y$ by the equivalence relation generated by $a \sim f(a)$ for all $a \in A$. Such a quotient space is called an ***adjunction space***, and is said to be formed by ***attaching Y to X along the attaching map f***.

Note that this equivalence relation identifies each point $y \in Y$ with the entire preimage $f^{-1}(y) \subseteq A$. If $A = \emptyset$, then our relation is empty and we get $Y \cup_f X = X \amalg Y$. If $Y = \{y\}$ only contains a single point, we get $Y \cup_f X = X/A$.

Theorem 6.3.2. Let $Y \cup_f X$ be an adjunction space. Let

$$q : X \amalg Y \to Y \cup_f X$$

be the associate quotient map. Then:

1. $q|_Y : Y \to Y \cup_f X$ is an embedding onto a closed subspace,
2. $q|_{X \setminus A} : X \setminus A \to Y \cup_f X$ is an embedding onto an open subspace,
3. $Y \cup_f X = q(Y) \amalg q(X \setminus A)$.

Proof. 1. Let $B \subseteq Y$ be closed. To show that $q(B) \subseteq Y \cup_f X$ is closed, we need to show that $q^{-1}(q(B))$ is closed in $X \amalg Y$, i.e. that $q^{-1}(q(B)) \cap X$ is closed in X and $q^{-1}(q(B)) \cap Y$ is closed in Y.

Since no two distinct elements of Y can be related to each other, we get $q(B) = B$, and since every element of Y is identified with itself and with its preimages, we get $q^{-1}(q(B)) = B \cup f^{-1}(B)$. Intersecting with X and Y gives

$$q^{-1}(q(B)) \cap Y = B$$
$$q^{-1}(q(B)) \cap X = f^{-1}(B).$$

The former is closed by definition. It follows that $f^{-1}(B)$ is closed in A, and since A is closed in X so is $f^{-1}(B)$. $q|_Y$ is clearly injective, since no points in Y are identified with each other. Thus, $q|_Y$ is a continuous closed injection, and thus an embedding. Note that $q|_Y$ being closed implies that $q(Y)$ is closed.

2. Every element of $X \setminus A$ is exclusively related to itself, so $X \setminus A$ is saturated. Therefore, $q|_{X \setminus A}$ is a quotient map onto its image by Theorem 6.2.11. Since every equivalence class contains only one element, this map is additionally injective, and since $q^{-1}(q(x)) = x$ for all $x \in X \setminus A$, this map is clopen, and thus an embedding.

□

Theorem 6.3.3. If $f : X \to Y$ is a quotient map and K is a locally compact Hausdorff space, then the function $f \times \mathrm{id}_K : X \times K \to Y \times K$ is a quotient map.

Definition 6.3.4. Let $f : X \to Y$ be a continuous function. Then the ***mapping cylinder*** of f is the space

$$M_f = Y \cup_{f_0} (X \times [0, 1]),$$

where $f_0 : X \times \{0\} \to Y$ is the function $f_0(x, 0) = f(x)$.

By abuse of notation, we identify the subset $X \times \{1\} \subseteq M_f$ with X itself.

Definition 6.3.5. Let $f : X \to Y$ be a continuous function. Then the ***mapping cone*** of f is the space

$$C_f = M_f/(X \times \{1\})$$

Theorem 6.3.6. A function $M_f \to Z$ is continuous if and only if the induced functions $X \times I \to Z$ and $Y \to Z$ are both continuous.

Chapter 7

Topological Manifolds

Definition 7.0.1

We call a topological space X ***locally Euclidean of dimension n*** if every point $x \in X$ has a neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Exercise 7.0.2. *If a topolocical space X is a locally euclidean of dimension n, then:*

1. *Every point $x \in X$ has a neighbourhood homeomorphic to an open ball in $\mathbb{R}^n$, or*
2. *Every point $x \in X$ has a neighbourhood homeomorphic to $\mathbb{R}^n$, or*

Exercise 7.0.3. *Show that a topological space is locally Euclidean of dimension 0 if and only if it is discrete.*

Definition 7.0.4

A topological space is called an ***n-dimensional topological manifold***, or just ***topological n-manifold***, if it is:

1. locally euclidean of dimension n,
2. second countable,
3. and Hausdorff.

The latter two requirements exclude some pathological spaces - the long line would be a topological manifold if it was second countable, and the bug-eyed line would be a topological manifold if it was Hausdorff. Most interesting results about topological manifolds require both of these properties, so little would be gained by omitting them.
Intuitively, it sounds very sensible that a nonempty topological space cannot be both a topological m-manifold and a topological n-manifold. This statement turns out to be true - however, we will save this proof for later, since it requires a substantial amount of machinery borrowed from algebraic topology.

Lemma 7.0.5. Every open subset of a topological n-manifold is a topological n-manifold.

Exercise 7.0.6. *A topological space is a topological 0-manifold if and only if it is countable and discrete.*

Theorem 7.0.7. Any separable metric space that is locally Euclidean of dimension n is a topological n-manifold.

Definition 7.0.8

We call a topological space an ***n-dimensional manifold with boundary*** if it is second countable, Hausdorff, and every open point has a neighbourhood homeomorphic to an open subset of either $\mathbb{R}^n$, or to the n-dimensional upper half-space

$$\mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$$

Definition 7.0.9

Let M be a topological n-manifold with boundary. Then a ***coordinate chart for M*** is a pair (U, φ), consisting of an open subset $U \subseteq M$, called a ***coordinate domain***, and a homeomorphism φ from U to an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, called a ***coordinate map***. If U is homeomorphic to an open ball, we call U a ***coordinate ball***.

Note that if we can assign coordinate maps for some covering of a topological space X, then that space must by definition be locally euclidean.

Definition 7.0.10: Manifold interior and manifold boundary

If $\varphi(U)$ is an open subset of $\mathbb{R}^n$, we call φ an ***interior chart*** and points $p \in U$ ***interior points***. If $\varphi(U)$ is a subset of $\mathbb{H}^n$ with $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$, we call φ a ***boundary chart***. If there exists a boundary chart taking p to $\partial\mathbb{H}^n$, we call p a ***boundary point of M***.

Every manifold is a manifold with boundary, but the opposite might not hold. We sometimes use the redundant term "manifold without boundary" to refer to a manifold with boundary which has the empty set as its boundary. The word "manifold", without further qualifiers, always means a manifold without boundary.
The definition of a manifold's boundary only agrees with the topological definition once we find the right ambient space to embed M into, since M always has empty topological boundary when viewed as a topological space of its own.
For example, the manifold boundary of the closed unit disk $\overline{\mathbb{B}}^2$ is the unit circle, which agrees with the topological boundary if the disk is viewed as a subset of $\mathbb{R}^2$, but not with the topological boundary of the disk when viewed as its own topological space (which is trivially empty, as stated), or with the topological boundary of the disk viewed as a subset of $\mathbb{R}^3$ (which is the entire disk).

Theorem 7.0.11. If M is a topological n-manifold with boundary, then $\operatorname{int} M$ is both an open subset of M and a topological n-manifold without boundary.

Proof. Since $\operatorname{int} M$ is a union of coordinate domains, it is open. Second-countability and Hausdorffness are trivially inherited by subspaces. By definition, $\operatorname{int} M$ consists only of points homeomorphic to an open subset of $\mathbb{R}^n$, so $\operatorname{int} M$ is additionally locally euclidean of dimension n. □

Proposition 7.0.12. Let M be a topological manifold with boundary. Then a point $x \in M$ cannot be both a boundary point and an interior point.

The general proof once again requires some tricky algebraic topology, so we save it for later.

Corollary 7.0.13. If M is a nonempty topological n-manifold with boundary, then ∂M is a closed subset of M, and M is a topological n-manifold if and only if $\partial M = \emptyset$.

Proof. Since $\operatorname{int} M$ is an open subset of M, $\partial M = M \setminus \operatorname{int} M$ is closed. If M is a manifold, then every point is an interior point, so $M = \int M$, so $\partial M = \emptyset$, and vice versa. □

7.1 Manifolds from Embeddings

Theorem 7.1.1

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is a topological n-manifold.

Proof. By 4.3.17, $\Gamma(f)$ is homeomorphic to $U \subseteq \mathbb{R}^n$, making $\Gamma(f)$ a trivial topological n-manifold coverable by a single coordinate domain. □

Theorem 7.1.2

The unit n-spheres $\mathbb{S}^n$ are topological n-manifolds.

Proof. For each $i \in [1...n+1]$, let U_i^+ and U_i^- denote the sets

$$U_i^+ := \left\{x \in \mathbb{R}^{n+1} \;\middle|\; x_i > 0\right\}$$
$$U_i^- := \left\{x \in \mathbb{R}^{n+1} \;\middle|\; x_i < 0\right\},$$

which are open in $\mathbb{R}^{n+1}$.
Since $\mathbb{S}^n$ doesn't contain the origin, every point on $\mathbb{S}^n$ must have at least one non-zero coordinate, so these sets cover $\mathbb{S}^{n+1}$.
On these sets, $\|x\| = 1$ has a unique solution, so we can find the intersection $x \in \mathbb{S}^n \cap U_i^\pm$ by solving for x_i, which gives us

$$\begin{aligned}
&\sqrt{\sum_{j=1}^{n+1} x_j^2} = 1 \\
\Leftrightarrow &\sum_{j=1}^{n+1} x_j^2 = 1 \\
\Leftrightarrow &\; x_i^2 = 1 - \sum_{j \neq i} x_j^2 \\
\Leftrightarrow &\; x_i = \begin{cases} \sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^+ \\ -\sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^- \end{cases}
\end{aligned}$$

We can thus describe the intersections of $\mathbb{S}^n$ with our sets $U_i^\pm$ by taking every component except the i-th to be arbitrary, and setting the i-th component to the value taken by these continuous square root functions. Swapping the i-th and the $n+1$-st components is continuous, so our intersections are the graphs of these functions. Since these intersections cover $\mathbb{S}^n$, and our functions are $\mathbb{R}^n \to \mathbb{R}$, $\mathbb{S}^n$ is a topological n-manifold.
□

Theorem 7.1.3. Let $p \in \mathbb{S}^n$. Let $H := \langle p \rangle^\perp$ be the hyperplane given as the orthogonal complement of the line through p and the origin. Then the ***stereographic projection map*** σ_p taking a point $x \in \mathbb{S}^n \setminus \{p\}$ to the point where the line $\vec{px}$ intersects H, is a homeomorphism $\mathbb{S}^n \setminus \{p\} \cong H (\cong \mathbb{R}^n)$

Proof. Let $N := e_{n+1} = (0, \ldots, n+1)^\top$ be the "north pole" of the sphere. Take a rotation R in the plane spanned by p and N mapping p to N. Since rotations in the origin preserve $\mathbb{S}^n$, we get

$$\sigma_p = R^{-1} \circ \sigma_N \circ R.$$

Since rotations are orthogonal maps, and thus homeomorphisms, it suffices to show that σ_N is a homeomorphism.

In this case, the hyperplane H orthogonal to $\langle N \rangle$ is simply given by $x_{n+1} = 0$. Therefore, the intersection $u := \sigma_N(x)$ of the line $\vec{Nx} = \{\lambda.(x - N) + N \mid \lambda \in \mathbb{R}\}$ can be found by finding the value of λ that makes the $n+1$st component zero. We get:

$$\begin{aligned} &\lambda.(x_{n+1} - 1) + 1 = 0 \\ \Leftrightarrow\ &\lambda.(x_{n+1} - 1) = -1 \\ \Leftrightarrow\ &\lambda = \frac{1}{1 - x_{n+1}} \end{aligned}$$

For $i \neq n+1$, we have $N_i = 0$, so $u_i = \lambda.x_i$. Substituting our λ into this equation gives

$$\sigma_N(x) = \frac{x_1, \ldots, x_n, 0}{1 - x_{n+1}}$$

This map is continuous for $x_{n+1} \neq 1$, i.e. $x \in \mathbb{S}^n \setminus \{N\}$.
Solving our equations for x instead gives

$$\begin{aligned} x_i &= \frac{u_i}{\lambda}, \\ x_{n+1} &= \frac{\lambda - 1}{\lambda}, \end{aligned}$$

And since our preimage needs to lie on $\mathbb{S}^n$, we substitute into $\|x\|^2 = 1$

and get

$$\begin{aligned}
1 &= \sum_{i=1}^{n+1} x_i^2 \\
&= \left(\sum_{i=1}^{n} \left(\frac{u_i}{\lambda}\right)^2\right) + \left(\frac{\lambda - 1}{\lambda}\right)^2 \\
&= \frac{\sum_{i=1}^{n} u_i^2}{\lambda^2} + \frac{\lambda^2 - 2\lambda + 1}{\lambda^2} \\
&= \frac{\|u\|^2 + \lambda^2 - 2\lambda + 1}{\lambda^2} \\
\Leftrightarrow \lambda^2 &= \|u\|^2 + \lambda^2 - 2\lambda + 1 \\
\Leftrightarrow 2\lambda &= \|u\|^2 + 1 \\
\Leftrightarrow \lambda &= \frac{\|u\|^2 + 1}{2}
\end{aligned}$$

Substituting back into our two equations gives

$$\sigma_N^{-1}(u_1, \ldots, u_n) = \frac{(2u_1, \ldots, 2u_n, \|u\|^2 - 1)^\top}{\|u\|^2 + 1},$$

which is continuous. Therefore, $\sigma_N : \mathbb{S}^n \setminus \{N\} \cong H \cong \mathbb{R}^n$ is indeed a homeomorphism, implying that every σ_p is a homeomorphism. □

σ_N and σ_S (where $S = -N$ is the "south pole") give an alternate coordinate atlas on $\mathbb{S}^n$, which is sometimes easier to work with than our original atlas given by square root functions.

Chapter 8

Topologies on Algebraic Structures

8.1 Topological Groups

Definition 8.1.1

Let $(G, \circ)$ be a group. Let $\mathcal{O}$ be a topology on G. Then we call $(G, \circ, \mathcal{O})$ a ***topological group*** iff. the maps $^{-1} : g \to g^{-1}$ and $\circ : G \times G \to G$ are continuous.

Exercise 8.1.2. *Show that $(G, \circ, \mathcal{O})$ is a topological group if and only if the map*

$$\varphi : (g, h) \mapsto g \circ h^{-1}$$

is continuous.

Exercise 8.1.3. *Let $(G, \circ, \mathcal{O})$ be a topological groups. Then for any $g \in G$, the left and right group operation maps $l_g : h \mapsto g \circ h$ and $r_g : h \mapsto h \circ g$ are homeomorphisms.*

Exercise 8.1.4. *Let $(G, \circ, \mathcal{O}_G)$ be a topological group. Let $(H, \circ, \mathcal{O}_H)$ be a normal subgroup equipped with the supspace topology, and let $(G/H, \circ, \mathcal{O}_q)$ be the quotient group, equipped with the quotient topology. Then the quotient map $p : G \to G/H$ is continuous and open.*

Corollary 8.1.5. If H is closed in G and G is Hausdorff, then G/H is Hausdorff. If G/H is T1, then H is closed.

Chapter 9

Connectedness

Theorem 9.0.1. Let $\{C_i\}_{i\in I}$ be a collection of connected sets in a topological space X such that no two of the C_i are disjoint. Then $\bigcup_{i\in I} C_i$ is connected.

Theorem 9.0.2. Let C be connected. Then every set D with $C \subseteq D \subseteq \overline{C}$ is connected. In particular, $\overline{C}$ is connected.

> *Proof.* Assume that D isn't connected. Then we can write $D = U \cap V$, with U and V disjoint and open. Since $C \subseteq D$, we then have $C = U' \cap V'$ for $U' = U \cap C$ and $V' = V \cap C$. Both sets are open in C by definition of the subspace topology, and since every neighbourhood of a point in $D \subseteq \overline{C}$ contains a point in C, neither of the two are empty. Therefore, C cannot be connected. □

Corollary 9.0.3. Connected components of a topological space are closed.

Chapter 10

Filters

10.1 Filter Convergence

It would be nice to find a generalization of sequence convergence to non-first-countable spaces that shares many of its useful properties.... It turns out that we can! The correct objects for the task are ***filters***.

Exercise 10.1.1. *Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every convergent filter on X has a unique limit.*

Theorem 10.1.2. Let X be a topological space. Let $U \subseteq X$. Then U is open iff. U is contained in every ultrafilter converging to some point in U.

Proof. $\Rightarrow$: By definition, if U is an open neighborhood of x, any filter convering to x must contain U.

$\Leftarrow$: Let $U \subseteq X$ and assume $U \in \mathcal{U}$ for every ultrafilter converging to a point $u \in U$. Assume U is not open. Then there exists a point $u \in U$ such that every open neighborhood of u contains at least one point in $X \setminus U$. But then any two members of the family

$$\{X \setminus U\} \cup \{U(u)\}$$

have nonempty intersection. Therefore, this family can be extended to an Ultrafilter $\mathcal{U}_u$. Since $\mathcal{U}_u$ contains all open neighborhoods of u by construction, it converges to u, and must thus contain U. Since it also contains $X \setminus U$, we arrive at a contradiction.

□

10.2 Tychonoff's Theorem

Definition 10.2.1. Let $\mathcal{F}$ be a filter on X. Let $f : X \to Y$. Then the ***pushforward Filter of f on $\mathcal{F}$*** is the set

$$f_*(\mathcal{F}) := \left\{ A \subseteq Y \;\middle|\; p_i^{-1}[A] \in \mathcal{F} \right\},$$

Exercise 10.2.2. *Verify that the pushforward of a filter is a filter, and that the pushforward of an ultrafilter is an ultrafilter.*

Intuitively, the pushforward of a filter under a function generalizes the sequence $f(x_n)$ for some sequence x_n and function f. By extension, the following theorem generalizes the fact that in metric spaces, continuity and sequential continuity are equivalent:

Theorem 10.2.3: Characterization of continuity in terms of ultrafilters

Let $f : X \to Y$. Then f is continuous if and only if $f_*(\mathcal{U})$ converges to $f(x)$ for every ultrafilter $\mathcal{U}$ that converges to $x \in X$.

Proof. 1. Assume f is continuous. Let $\mathcal{U}$ be an ultrafilter on X converging to $x \in X$.

Let $U \subseteq Y$ be an open neighbourhood of $f(x)$. Then, since f is continuous, $f^{-1}(U)$ is an open neighbourhood of x. We thus have $f^{-1}(U) \in \mathcal{U}$, which gives us $U \in f_*(\mathcal{U})$ by the definition of the pushforward filter.

Thus, every open neighbourhood of $f(x)$ is contained in $f_*(\mathcal{U})$, so $f_*(\mathcal{U})$ converges to $f(x)$.

2. Assume that the pushforward of any ultrafilter converging to x converges to $f(x)$. Let $U \subseteq Y$ be open.

Let $\mathcal{U}$ be an ultrafilter on X converging to some point $x \in f^{-1}(U)$. Then, by our hypothesis, $f_*(\mathcal{U})$ converges to $f(x)$. Since U is an open neighborhood of $f(x)$, we have $U \in f_*(\mathcal{U})$, and thus $f^{-1}(U) \in \mathcal{U}$. Thus, by theorem 10.1.2, $f^{-1}(U)$ is open, so f is continuous.

□

This theorem generalizes the fact that a sequence in a metric product space converges if and only if every individual component converges.

Theorem 10.2.4. Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Let $\mathcal{F}$ be a filter on $\prod_{i \in I} X_i$. Then $\mathcal{F}$ converges to a point $x \in \prod_{i \in I} X_i$ if and only if each pushforward filter $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$.

Proof. $\Rightarrow$: Assume that $\mathcal{F}$ converges to $x \in \prod_{i \in I} X_i$. Since all projections p_i are continuous, each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$ by theorem 10.2.3.

$\Leftarrow$: Assume that each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$. By definition of the product topology, the sets $S_i := p_i^{-1}(U_i)$ for open $U_i \in X_i$ form a subbase of $\prod_{i \in I} X_i$. Therefore, any open neighborhood $U \in \mathcal{U}(x)$ contains an open neighborhood $B = \bigcap_{i \in J \subseteq I} p_i^{-1}(U_i)$ for some open neighborhoods $U_i \in \mathcal{U}(x_i)$.

Since each $(p_i)_*(\mathcal{F})$ converges to x_i, we have $U_i \in (p_i)_*(\mathcal{F})$ for each i. Therefore, by definition of the pushforward filter, we have $p_i^{-1}(U_i) \in \mathcal{F}$ for all i. Since filters are closed under intersection, we get $B \in \mathcal{F}$,

and since $B \subseteq U$ we also get $U \in \mathcal{F}$ as desired.

□

Theorem 10.2.5: Tychonoff's Theorem

Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Then if all X_i are compact, $\prod_{i \in I} X$ is compact.

Proof. 1. If there exists an $i \in I$ such that $X_i = \emptyset$, then $\prod X_i = \emptyset$ is compact.

2. Now, assume $\prod_{i \in I} X_i \neq \emptyset$. Then there exists an ultrafilter $\mathcal{U}$ on $\prod_{i \in I} X_i$.

   Since each X_i is compact, each pushforward $(p_i)_*(\mathcal{U})$ converges to an $x_i \in X_i$. Thus, by theorem 10.2.4, $\mathcal{U}$ converges to $(x_i)_{i \in I}$ and $\prod_{i \in I} X_i$ is compact.

□

Chapter 11

Higher Separation Properties

11.1 Regular Spaces

Definition 11.1.1: Regular Space

Let (X, O) be a topological space. Then we call (X, O) ***regular*** if every point can be housed orf from any closed set not containing it:

$$\begin{aligned}&\forall (X \setminus C) \in O : \\ &\forall x \notin C : \\ &\exists U_x \in \mathcal{U}(x), U_C \in \mathcal{U}(C) : \\ &\quad U_x \cap U_C = \emptyset\end{aligned}$$

We call (X, O) ***T3*** or ***regular Hausdorff***, if it is regular and Hausdorff.

Exercise 11.1.2. *Find a topological space that is regular, but not To.*

Exercise 11.1.3. *Show that a topological space is regular Hausdorff (T_3) if and only if it is regular and T_1.*

Theorem 11.1.4. Every compact Hausdorff space is regular.

Proof. Let $C \subseteq X$ be closed, and let $y \in X \setminus C$. Since X is Hausdorff, there exist disjoint open neighbourhoods $U_x \ni x \in C$ and $V_x \ni y$ for each $x \in C$. Let $\mathcal{U} := \{U_x \mid x \in C\}$. Since C is a closed subset of the compact space X, C is also compact. Therefore, the open cover $\mathcal{U}$ of C has a finite subcover

$$\mathcal{U}' = \left\{U_{x_i} \,\middle|\, i \in [1..n]\right\}.$$

If we now set

$$\mathcal{V}' = \left\{V_{x_i} \,\middle|\, i \in [1..n]\right\},$$

we have our desired separation of x and C via

$$\begin{aligned}U &= \bigcup \mathcal{U}' \\ V &= \bigcap \mathcal{V}'.\end{aligned}$$

□

Theorem 11.1.5. A Hausdorff space is regular if and only if the closed neighbourhoods of any point form a neighbourhood basis of that point.

Definition 11.1.6: Completely Regular Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***completely regular*** if every point can be housed orf from any closed set not containing it by a continuous function:

$$\begin{aligned}&\forall (X \setminus C) \in \mathcal{O} : \\ &\forall x \notin C : \\ &\exists f \in C(X, [0,1]_{\mathbb{R}}) : \\ &\quad f(x) = 1,\ f[A] \subseteq \{0\}\end{aligned}$$

We call $(X, \mathcal{O})$ ***T3a***, ***T3$\frac{1}{2}$***, or ***completely regular Hausdorff***, if it is completely regular and Hausdorff.

Exercise 11.1.7. *Show that every completely regular space is regular.*

Exercise 11.1.8. *Find a space that is completely regular, but not T0.*

Exercise 11.1.9. *Show that a space if completely regular Hausdorff if and only if it is completely regular and T1.*

Theorem 11.1.10. Every zero-dimensional space $(X, \mathcal{O})$ is completely regular.

Proof. Since X is zero-dimensional, it has a basis consisting of clopen sets. Given any clopen set U, the function

$$f_U : x \mapsto \begin{cases} 1 & x \in U \\ 0 & x \in X \setminus U \end{cases}$$

is continuous. Now, let C be a closed set not containing x. Then $X \setminus C$ is an open neighbourhood of x. We can now choose a basic (i.e. clopen) neighbourhood $U \subseteq X \setminus C$ of x, at which point f_U is our desired function separating x and C. □

Theorem 11.1.11. Let $(X, \mathcal{O})$ be completely regular. Then the set

$$\left\{ f^{-1}[U] \,\middle|\, f \in C(X, [0,1]), U \in \mathcal{O}_{\mathbb{R}} \right\}$$

is a basis of X.

Proof.

1. Since the f are continuous, every basis set is open.
2. Let $x \in X$. Let U be an open neighbourhood of x. Since X is completely regular, there exists a continuous function f separating $\{x\}$ and $X \setminus U$. Then the basic set $f^{-1}[(0,1]]$ must be a subset of U, since every element outside of U gets mapped to 0. Thus, every open neighhourhood contains a basic neighbourhood.

□

Theorem 11.1.12. Every completely regular Hausdorff space $(X, \mathcal{O})$ can be embedded into the cube

$$\prod_{f \in C(X,[0,1])} [0,1]$$

Proof. We choose the trivial embedding $\iota : x \mapsto (f(x))_{f \in C(X,[0,1])}$.

1. ι is continuous, since the projections of ι are exactly the elements of $C(X, [0,1])$, i.e. continuous.

2. Let $x \neq y$. By T1, all singletons are closed. Therefore, by complete regularity, there exists a function f with $f(x) = 1$ and $f(y) = 0$. Therefore, we have $\iota(x)_f \neq \iota(y)_f$, i.e. $\iota(x) \neq \iota(y)$.

3. By 11.1.11, we only need to show openness for the basis sets

$$\left\{ f^{-1}[U] \,\middle|\, f \in C((X, \mathcal{O}), [0,1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}} \right\}.$$

   For these sets, we get:

$$\begin{aligned}
\iota\left[f^{-1}[U]\right] &= \iota[(p_f \circ \iota)^{-1}[U]] \\
&= \iota[\iota^{-1}(p_f^{-1}[U])] \\
&= \iota\left[\iota^{-1}\left[U \times \prod_{g \neq f} [0,1]\right]\right] \\
&= \left(U \times \prod_{g \neq f} [0,1]\right) \cap \iota[X]
\end{aligned}$$

   Where $\left(U \times \prod_{g \neq f} [0,1]\right)$ is open, since every component besides the f-component is open.

□

11.2 Normal Spaces

Definition 11.2.1: Normal Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***normal*** if every pair of disjoint closed sets can be housed orf from each other by disjoint neighbourhoods:

$$\begin{aligned} &\forall X \setminus C_1, X \setminus C_2 \in \mathcal{O} : \\ &\quad C_1 \cap C_2 = \emptyset \implies \\ &\exists U_1 \in \mathcal{U}(C_1), U_2 \in \mathcal{U}(C_2) : \\ &\quad U_1 \cap U_2 = \emptyset \end{aligned}$$

We call $(X, \mathcal{O})$ ***T4***, or ***normal Hausdorff***, if it is normal and Hausdorff.

Exercise 11.2.2. *Find a topological space that is normal, but not To.*

Exercise 11.2.3. *Show that every compact Hausdorff space is normal. (Hint: Our proof of Theorem 11.1.4 (every compact Hausdorff space is regular) gets us most of the way there and only needs a small alteration.)*

Lemma 11.2.4: Urysohn's Lemma

A topological space is normal if and only if any two disjoint closed subsets can be separated by a continuous function.

Corollary 11.2.5. Every normal Hausdorff space is completely regular Hausdorff.

Theorem 11.2.6. A second-countable regular topological space is normal.

Proof. Let A, B be disjoint closed sets in a second-countable regular topological space X. Let $\{U_i\}_{i \in I}$ be the collection of open sets such that $U \cap A \neq \emptyset$ and $\overline{U} \cap B = \emptyset$. Since X is regular, this collection forms an open covering of A, and X is second-countable, we can take a countable subcovering $\{U_n\}_{n \in \mathbb{N}}$.
Analogously, we can construct a countable open covering $\{V_n\}_{n \in \mathbb{N}}$ of B fulfilling $\overline{V_n} \cap A = \emptyset$.
Now, in order to make these two coverings disjoint from each other as well, we take

$$Y_n := U_n \cap \bigcap_{k=1}^{n} X \setminus \overline{V}_k$$

$$Z_n := V_n \cap \bigcap_{k=1}^{n} X \setminus \overline{U}_k$$

By construction of our U_n and V_n, our Y_n form an open cover of A and our Z_n form an open cover of B. Furthermore, we have $Y_m \cap Z_n = \emptyset$ for

all $m, n \in \mathbb{N}$, since $m \geq n$ implies

$$\begin{aligned} Y_m \cap Z_n &\supseteq Y_m \cap V_n \\ &\supseteq Y_m \cap \overline{V}_n \\ &\supseteq U_m \cap (X \setminus \overline{V}_n) \cap \overline{V}_n \\ &= \emptyset, \end{aligned}$$

and $n \geq m$ implies the same, but with Y, Z and U, V swapped.
Therefore, $\bigcup_{m\in\mathbb{N}} Y_m$ and $\bigcup_{n\in\mathbb{N}} Z_n$ are disjoint open sets containing A and B, completing the proof that X is normal. □

11.3 Urysohn's Metrization Theorem

Corollary 11.3.1: Urysohn's Metrization Theorem

Every second-countable completely regular Hausdorff space X can be embedded into the Hilbert Cube

$$\prod_{n\in\mathbb{N}}[0,1]$$

Proof. Since X is second-countable completely regular Hausdorff, X is normal Hausdorff by Theorem 11.2.6. Now, we define a set

$$S = \left\{(U,V) \in \mathcal{B}^2 \;\middle|\; \overline{U} \subseteq V\right\}.$$

Urysohn's Lemma 11.2.4 tells us that for each $s = (U,V) \in S$, there exists a continuous function $g_s : X \to [0,1]$ such that $g_s[\overline{U}] = 1$ and $g_s[X \setminus V] = 0$. This lets us find a canonical continuous map $X \to \prod_{s\in S}[0,1]$:

$$\begin{aligned} g : X &\to \prod_{s\in S}[0,1] \\ x &\mapsto (s \mapsto g_s(x)) \end{aligned}$$

This map is injective, since for $x \neq y$, we can use complete regularity to find a pair $s = (U,V) \in S$ with $x \in U$ and $y \notin V$ such that $g_s(x) = 1$ and $g_s(y) \in g_s[X \setminus V] = 0$, i.e. $g(x) \neq g(y)$.

It remains to be shown that g is open. If W is an open neighbourhood of $x \in X$, we want the set $g(W)$ to be open in $\prod_{s\in S}[0,1]$, which follows if we can find for each $g(y) \in g(W)$ a subbasis neighbourhood $\pi^{-1}(S)$ of the product topology such that

$$g(y) \in g(X) \cap \pi^{-1}(S) \subseteq g(W)$$

Now, as earlier, we find basic neighbourhoods $V \subseteq W \in \mathcal{B}$ and $s = (U,V) \in S$ such that $g_s(x) \in g_s[\overline{U}] = 1$ and $g_s[X \setminus V] = 0$, i.e.

$$g_s(y) \neq 0 \implies y \in V,$$

i.e. $g_s^{-1}[(0,1]] \subseteq V$. Since we have $g_s = \pi_s \circ g$, we get $g_s^{-1} = g^{-1} \circ \pi_s^{-1}$. Fiddling around with images and preimages for a bit now gives us:

$$\begin{aligned} g_s^{-1}[(0,1]] \cap X = g^{-1}(\pi_s^{-1}[(0,1]]) \cap X \subseteq V \\ &\implies g(g^{-1}(\pi_s^{-1}[(0,1]]) \cap X) \subseteq g[V] \\ &\overset{g \text{ inj.}}{\implies} g(g^{-1}(\pi_s^{-1}[(0,1]])) \cap g[X] \subseteq g[V] \\ &\implies \pi_s^{-1}[(0,1]] \cap g[X] \subseteq g[V] \end{aligned}$$

□

11.4 Local Compactness and Compactification

Definition 11.4.1

We call a topological space ***locally compact*** if every point has a compact neighbourhood.

Theorem 11.4.2. Let X be a locally compact Hausdorff space. Then:

1. X is completely regular.
2. Every neighbourhood of $x \in X$ contains a compact neighbourhood of x.

Proof. Let C be a compact neighbourhood of x. Let U be an arbitrary neighbourhood of x. Let $V \subseteq C \cap U$ be open with $x \in V$. Then $\overline{V} \subseteq C$ is a compact Hausdorff space, and thus normal by Exercise 11.2.3 and completely regular by Urysohn's Lemma 11.2.4. Since every point has a completely regular neighbourhood, X is completely regular.

Since $V \subseteq \overline{V}$ is also completely regular, the closed neighbourhoods of any point $x \in V$ form a neighbourhood basis by Theorem 11.1.5.

Thus, there exists a neighbourhood $N \subseteq V$ of $x \in C$ which is closed in $\overline{V}$, and thus closed in X. Since $N \subseteq C$ is closed and C is compact, N is also compact. Since N is a neighbourhood of x in $\overline{V}$ and $N \subseteq V$,< N is also a neighbourhood of x in the open set V, and thus in all of X. □

Theorem 11.4.3: One-Point Compactification

Let $(X, \mathcal{O})$ be a locally compact Hausdorff space. Let $X^+ = X \cup \{\infty\}$ be the set consisting of X and an additional "point at infinity" not originally contained in X.

Define the collection of open sets in X^+ to be

$$\mathcal{O}^+ := \mathcal{O} \cup \left\{ X^+ \setminus C \;\middle|\; C \subseteq X \text{ compact} \right\},$$

i.e. the original open sets in X plus the complements of compact sets in X with the additional point ∞ added. Then $(X^+, \mathcal{O}^+)$ is a compact Hausdorff space called the ***one point compactification*** of X. Furthermore, $\mathcal{O}^+$ is the only topology making the set X^+ a compact Hausdorff space with X as a subspace.

11.5 Paracompactness

Definition 11.5.1. Let $\mathcal{U}$ and $\mathcal{V}$ be coverings of a topological space X. We call $\mathcal{U}$ a ***refinement*** of $\mathcal{V}$ if each element of $\mathcal{U}$ is a subset of some element of $\mathcal{V}$.

Definition 11.5.2. Let $\mathcal{U}$ be a collection of subsets of a topological space X. We call $\mathcal{U}$ ***locally finite*** if each point $x \in X$ has a neighbourhood N which only intersects a finite number of elements of $\mathcal{U}$.

Theorem 11.5.3. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be a locally finite collection of subsets of a topological space X. Then

$$\overline{\bigcup_{i\in I} U_i} = \bigcup_{i\in I} \overline{U}_i$$

Proof. $\supseteq$: Follows trivially, since

$$\begin{aligned} &\forall i \in I : \bigcup_{i\in I} U_i \supseteq U_i \\ \implies &\forall i \in I : \overline{\bigcup_{i\in I} U_i} \supseteq \overline{U}_i \\ \implies &\overline{\bigcup_{i\in I} U_i} \supseteq \bigcup_{i\in I} \overline{U}_i \end{aligned}$$

$\subseteq$: It suffices to show that $\bigcup_{i\in I} \overline{U}_i$ is closed: $\bigcup_{i\in I} \overline{U}_i$ trivially contains $\bigcup_{i\in I} U_i$, which means that $\overline{\bigcup_{i\in I} U_i}$ is contained in the closure of $\bigcup_{i\in I} \overline{U}_i$.

Thus, we pick $x \notin \bigcup_{i\in I} \overline{U}_i$. Local finiteness tells us that there is a neighbourhood $N \ni x$ which intersects only finitely many U_i, which we denote $U_{i_1}, \dots, U_{i_n}$. This lets us construct an open neighbourhood

$$N' = N \setminus \bigcup_{k=1}^{n} \overline{U}_{i_k},$$

which still contains x but doesn't intersect any U_i. Since N' is open, it also doesn't intersect any $\overline{U}_i$ (otherwise there would have to be a point $y \in N'$ which doesn't have a anyneighbourhood in N').

□

This equality doesn't hold for arbitrary collections, since an infinite union of closed sets may not be closed - consider for example $I = \mathbb{N}$ and

$$U_i := [1/n, 1 - 1/n].$$

Definition 11.5.4: Paracompact Spaces

We call a topological space X ***paracompact*** if every open covering of X has a locally finite open refinement.

Exercise 11.5.5. *Every closed subset of a paracompact space is paracompact.*

Theorem 11.5.6. Every paracompact Hausdorff space is regular.

Proof. We start by constructing disjoint neighbourhoods $U_x \ni x \in C$, $V_x \ni y \in X \setminus C$ as we did in the compact case 11.1.4. Then $\mathcal{U} = \{U_x \mid x \in C\}$ is an open covering of C. Let $\mathcal{U}' = \{U_i \mid i \in I\}$ be an open locally finite refinement, and let $U' = \bigcup_{i \in I} U_i$.

Since $\mathcal{U}'$ is locally finite, we have

$$\overline{U'} = \bigcup_{i \in I} \overline{U}_i$$

by Theorem 11.5.3.

Since y isn't contained in any U_x, and $y \in V_x$, which is an open subset disjoint from U_x, y isn't contained in any $\overline{U}_x$, and therefore can't be contained in any $\overline{U}_i \subseteq \overline{U}_x$ and especially not in their union $\bigcup_{i \in I} \overline{U}_i = \overline{U'}$.

Thus, $U' \supseteq C$ and $X \setminus \overline{U'} \ni y$ are disjoint open sets containing C and y respectively. □

Corollary 11.5.7. Every paracompact Hausdorff space is normal.

Proof. Let $C_1, C_2 \subseteq x$ be disjoint closed sets. Since every paracompact Hausdorff space is regular, we can apply the same construction as in the regular case, but set $x \in C_1$ and have each V_x be a neighbourhood of the entirety of C_2. □

Thus, every second countable paracompact space is metrizable by Urysohn's Lemma 11.2.4.

Proposition 11.5.8. Every metric space is paracompact.

Proof Sketch. The proof makes use of some set-theoretic trickery involving ordinals and the well-ordering theorem. If you're interested, a short version is given by Mary Ellen Rudin in *A new proof that metric spaces are paracompact*, AMS (1968) "□"

Definition 11.5.9

Let $f : X \to \mathbb{R}$ be a continuous map from a topological space into the real numbers. Then the ***support of f*** is the smallest closed set on which $f(x) \neq 0$:

$$\operatorname{supp} f = \overline{\left\{x \in X \;\middle|\; f(x) \neq 0\right\}}$$

Definition 11.5.10. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open covering of the space X. Then a ***partition of unity*** subordinate to this covering is a collection of maps

$$\left\{f_j : X \to [0,1] \;\middle|\; j \in J\right\}$$

such that:

1. There exists a locally finite open refinement $\left\{V_j \;\middle|\; j \in J\right\}$ of $\mathcal{U}$ such that

$$\forall j \in J : \operatorname{supp} f_j \subseteq V_j$$

2. The functions "partition unity" in the sense that they add up to the constant function $\mathbb{1}_X$:

$$\forall x \in X : \sum_{j \in J} f_j(x) = 1$$

Theorem 11.5.11. Let X be a paracompact topological space. Let $\mathcal{U}$ be an open covering of X. Then there exists a partition of unity subordinate to $\mathcal{U}$.

Corollary 11.5.12. Let X be paracompact. Let $\{U_i \mid i \in I\}$ be a locally finite open covering of X. Then there exists an open covering $\{V_i \mid i \in I\}$ such that for each $i \in I$, we have $\overline{V}_i \subseteq U_i$.

Definition 11.5.13. We call a topological space ***σ-compact*** if it is the union of countably many compact subspaces.

Theorem 11.5.14. A locally compact Hausdorff space is paracompact if and only if it is the disjoint union of open σ-compact subsets.

Theorem 11.5.15. Let X be a locally compact second countable Hausdorff space. Then X is σ-compact and paracompact.

In particular, X is a second countable completely regular Hausdorff space by Corollary 11.5.7, and thus metrizable by Urysohn's metrization theorem 11.3.1.

Proof. Let X be a locally compact second countable Hausdorff space. Let $\mathcal{B}$ be a countable basis of X. Let $C \subseteq X$ be compact. By local compactness, every point $x \in C$ has a compact neighbourhood N. Since N is a neighbourhood, there exists a basis element $U_x \in \mathcal{B}$ such that $x \in U_x \subseteq N$. Since $U_x \subseteq N$, each closure $\overline{U}_x$ is compact, so the closures of these basis elements give a countable covering of X by compact sets, i.e. X is σ-compact. By Theorem 11.5.14, X is also paracompact. □

Theorem 11.5.16. If X is a locally compact second countable Hausdorff space, then its one-point compactification X^+ is metrizable.

Proof. Continuing the construction from the previous proof, we can cover C by the our basis elements U_x and pick a finite subcover $\{U_{x_k} \mid k \in [1..n]\}$.

Then $\bigcup_{k=1}^{n} \overline{U_{x_k}}$ is a finite union of compact sets, i.e. compact, so the set

$$V_C = X^+ \setminus \bigcup_{k=1}^{n} \overline{U_{x_k}}$$

is an open neighbourhood of ∞ contained in an arbitrary neighbourhood of the form $X^+ \setminus C$.

Each set V_C is uniquely defined by a finite subset of the countable set $\mathcal{B}$, so there are only countably many V_C. Thus, $\mathcal{B} \cup \{V_C \mid C \text{ compact}\}$ is a countable basis of X^+. Furthermore, X^+ is a compact Hausdorff space by construction, and thus completely regular. Metrizability follows, as usual, from Urysohn's metrization theorem 11.3.1. □

Chapter 12

Lattice Theory

12.1 Lattices

Definition 12.1.1. A partially ordered set $(L, \leq)$ is called a ***lattice*** if every pair of elements $x, y \in L$ has a least upper bound $x \wedge y$, also known as a ***join***, and a greatest lower bound $x \vee y$, also known as a ***meet***. A ***complete lattice*** is a lattice in which every subset $A \subseteq L$ has a join $\sup A$ and a meet $\inf A$.

Definition 12.1.2. Given a complete lattice L, the ***bottom element*** is the element $0 := \inf L$, and the ***top element*** is the element $\infty := \sup L$.

Definition 12.1.3. Let L be a complete latice. Let $x, y \in L$. Then we say that x is ***well above*** y, denoted $x \succ y$, iff for any subset $A \subseteq L$,

$$y \geq \inf A \implies \exists a \in A : x \geq a$$

Lemma 12.1.4. Let L be a complete lattice. Let $x, y, z \in L$. Then:

1. $x \succ y$ implies $x \geq y$,
2. $x \succ y$ and $y \geq z$ implies $x \geq z$,
3. $x \geq y$ and $y \succ z$ implies $x \succ z$.

Proof.
1. We have $y \geq \inf \{y\}$, so $x \succ y$ implies $x \geq y$.
2. Follows from applying transitivity to the first part of the lemma.
3. $z \geq \inf A \implies \exists a \in A : x \geq y \geq a$.

□

Lemma 12.1.5. Let L be a complete lattice. Let $x \in L$ and $A \subseteq L$. Then

$$x \succ \inf A \Leftrightarrow \exists a \in A : x \succ a$$

12.2 Value Quantales

12.3 Generalized metrics metrize every topological space

Part III

Differential Geometry

Chapter 13

Smooth Manifolds

13.1 Abstract Smooth Manifolds

Definition 13.1.1: Smooth Manifolds via Charts

An ***n-dimensional smooth manifold*** is a second-countable Hausdorff space M together with a collection of maps called "charts" such that:

1. Each chart is a homeomorphism $\Phi : U \to U' \subseteq \mathbb{R}^n$, where U is open in M and U' is open in $\mathbb{R}^n$.
2. Each point $x \in M$ is in the domain of some chart,
3. Given charts $\Phi : U \to U'$ and $\Psi : V \to V'$, the coordinate change function $\Phi\Psi^{-1} : \Psi(U \cap V) \to \Phi(U \cap V)$ is smooth (i.e. infinitely continuously differentiable)
4. The collection of charts is maximal, i.e. if a map fulfills conditions (1) and doesn't violate condition (3) when taken together with preexisting charts, then that new map is a chart.

We call our any collection of charts satisfying conditions $1, 2$ and 3 an ***atlas*** of M.

Note that this definition directly generalizes topological manifolds, but with additional smoothness demands on our charts. Since most manifolds can't be equipped with the structure of a vector space, differentiability isn't well-defined for our charts $\Phi : U \to U \subseteq \mathbb{R}^n$, so instead we move that demand to the coordinate change functions $\Phi\Psi^{-1}$, which are functions mapping $\mathbb{R}^n$ into itself.

We also want to construct a second, slightly more abstract definition of a smooth manifold, based on the notion of ***functional structures***.

Definition 13.1.2

Let $(X, \mathcal{O})$ be a topological space. A ***fuctional structure*** on X is a function F_X defined on the collection $\mathcal{O}$ of open sets in X, such that:

1. $F_X(U)$ is a subalgebra of the algebra $C(U, \mathbb{R})$ of continuous real-valued functions on U[1],
2. $F_X(U)$ contains all constant functions,
3. If $V \subseteq U$ and $f \in F_X(U)$, then $f|_V \in F_X(V)$, and
4. If $U = \bigcup_{i \in I} U_i$ and $f|_{U_i} \in F_x(U_i)$ for all i, then $f \in F_X(U)$.

The pair (X, F) is called a ***functionally structured space***.

[1] Originally defined in Exercise **??**.

In particular, we care about the following functional structures on $X = \mathbb{R}^n$:

1. $F_X(U) = C(U, \mathbb{R})$,
2. $F_X(U) = C^k(U, \mathbb{R})$,
3. $F_X(U) = C^\infty(U, \mathbb{R})$,
4. $F_X(U) = A(U, \mathbb{R})$ (= the set of analytic functions $U \to \mathbb{R}$).

Exercise 13.1.3. *If (X, F_X) is a functionally structured space, we can turn (U, F_U) into a functionally structured space by setting $F_U(V) = F_X(V)$ for all subsets $V \subseteq U$.*

Definition 13.1.4

A ***morphism of functionally structured spaces***

$$(X, F_X) \to (Y, F_Y)$$

is a continuous function $\Phi : X \to Y$ such that precomposition pulls back elements of $F_Y(U)$ to elements of $F_X(\Phi^{-1}(U))$:

$$f \in F_Y(U) \implies f \circ \Phi \in F_X(\Phi^{-1}(U))$$

An isomorphism, as always, is a morphism whose inverse exists and is a morphism.

Definition 13.1.5: Smooth Manifolds via Functional Structure

An n-dimensional differentiable manifold is a second countable functionally structured Hausdorff space (M, F) which is locally isomorphic to $(\mathbb{R}^n, C^\infty)$, i.e. each point in M has a neighborhood U such that $(U, F_U) \cong (V, C^\infty_V)$.

Theorem 13.1.6. Our two definitions of smooth manifolds are equivalent.

Proof. $\Rightarrow$: Let M be a smooth manifold in the sense of Definition 13.1.1. We need to define the sets $F(U)$ for all open sets U. Let U be the domain of a chart $\Phi : U \to U' \subseteq \mathbb{R}^n$. Then we can set

$$F(U) = \left\{ f \circ \Phi \,\middle|\, f \in C^\infty(U', \mathbb{R}) \right\}$$

Then F inherits the properties defining a functional structure from $C^\infty(U', \mathbb{R})$, and Φ is an isomorphism between F and C^∞ by definition of F.

$\Leftarrow$: Let (M, F) be a smooth manifold in the sense of Definition 13.1.5. We take our charts $\Phi : U \to V$ to be isomorphisms $(U, F_U) \cong (V, C^\infty_V)$.

It remains to be shown that the coordinate change function $\theta := \Phi\Psi^{-1}$ of any pair of charts is C^∞, and that our atlas is maximal. Since θ is a morphism, $f \circ \theta$ is C^∞ for any $f \in C^\infty$. Therefore, $\pi_i \circ \theta$ is C^∞ for all $i \in [1..n]$, so θ is also C^∞.

At the same time, if Φ and Ψ are homeomorphisms and θ is C^∞, then θ is clearly an isomorphism, since $\theta \in C^\infty$ implies $f \circ \theta \in C^\infty$ for all $f \in C^\infty$ and $\theta^{-1} \in C^\infty$ exists and fulfills the same conditions. Therefore, our atlas is maximal.

□

Proposition 13.1.7. These two constructions are inverse to each other.

From now on, we will implicitly assume that every functional structure induced by a chart structure and every chart structure induced by a functional structure is created via this construction, where charts correspond to local isomorphisms of the functional structures.

This lets us generalize smoothness from vector spaces to manifolds:

Definition 13.1.8

A map between manifolds is called ***smooth*** or $\boldsymbol{C^\infty}$ if it is a morphism between their respective functional structures

Theorem 13.1.9

A map $f : M \to N$ between two smooth manifolds is smooth in the sense of being a morphism between functional structures if and only if, for any charts $\Phi : U \to U' \subseteq \mathbb{R}^m$ on M and $\Psi : V \to V' \subseteq \mathbb{R}^n$ on N, the function $\Psi \circ f \circ \Phi^{-1}$ is smooth.

Proof. $\Rightarrow$: Let $f : (M, F_M) \to (N, F_N)$ be a morphism. Let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ be the i-th coordinate projection function. Since charts are isomorphisms, we get:

$$\begin{aligned} & \pi_i \in C^\infty(\mathbb{R}^n, \mathbb{R}) \\ \implies & \pi_i \circ \Psi \in F_N(V) \\ \implies & \pi_i \circ \Psi \circ f \in F_M(f^{-1}(V)) \\ \implies & \pi_i \circ \Psi \circ f \circ \Phi^{-1} \in C^\infty(\Phi(f^{-1}(V)), \mathbb{R}) \end{aligned}$$

Since every component of $\Psi \circ f \circ \Phi^{-1}$ is smooth, the entire function is smooth.

$\Leftarrow$: Let $\Psi \circ f \circ \Phi^{-1}$ be smooth for all charts Φ, Ψ. Since Φ is bijective, the set on which this function is well-defined is

$$\Phi(f^{-1}(V)) \cap \Phi(U) = \Phi(f^{-1}(V) \cap U).$$

Let $g \in F_N(U)$. Then we have $g = h \circ \Psi$ for some chart $\Psi : V \to \mathbb{R}^n$ and some smooth real-valued function $h \in C^\infty(\mathbb{R}^n, \mathbb{R})$.

We get:

$$\begin{aligned} & g := h \circ \Phi \in F_N(V) \\ \implies & h \in C^\infty(\mathbb{R}^n, \mathbb{R}) \\ \implies & h \circ \Psi \circ f \circ \Phi^{-1} \in C^\infty(\Phi(f^{-1}(V) \cap U), \mathbb{R}) \\ \implies & (g \circ f)|_{f^{-1}(V) \cap U} = h \circ \Psi \circ f \circ \Phi^{-1} \circ \Phi \in F_M(f^{-1}(V) \cap U, \mathbb{R}) \end{aligned}$$

Picking different charts $\Phi_i : U_i \to \mathbb{R}^n$ whose domains cover $f^{-1}(V)$ gives $g \circ f \in F_M(f^{-1}(V))$ as desired.

□

Another concept that manifolds inherit from vector spaces is orientation:

Definition 13.1.10

Let M be an n-manifold. Then an atlas A which is maximal with regards to the property that, for any pair of charts Φ, Ψ in the atlas, the Jacobian of the coordinate change function $\Phi\Psi^{-1}$ has positive determinant at all points in its domain, is called an ***orientation*** of M. An n-manifold having such an atlas is called ***orientable***.

Theorem 13.1.11. Every orientable manifold has at least two orientations.

Proof Sketch. Given an orientation of M, we can pick one coordinate of each chart and reverse the sign, leading to a new orientation. "□"

Theorem 13.1.12. Every connected manifold has at most two orientations.

Proof. Since charts are bijective, their Jacobians never have determinant zero.

Now, $\mathcal{A}$ and $\mathcal{B}$ be two orientations. Define a map $s(p) : M \to \{1, -1\}$ by setting $s(p) = 1$ if all coordinate change functions $\Phi\Psi^{-1}$ involving one chart from each orientation have positive determinant (in which case all of them have positive determinant, since $\Phi\psi^{-1} = \Phi\Psi^{-1}\Psi\psi^{-1}$), and $s(p) = -1$ otherwise. Since Jacobian determinants vary continuously, $s(p)$ is locally constant. Since M is connected, $s(p)$ must be constant, leaving two options:

1. If $s(p) = 1$, then $\mathcal{A}$ and $\mathcal{B}$ are compatible, and must thus already contain each others charts, since orientations are maximal.
2. If $s(p) = -1$, let $\mathcal{C}$ be another orientation for which the analogously defined function s_C comparing $\mathcal{C}$ to $\mathcal{A}$ is negative. Then, given charts Φ_A, Φ_B, Φ_C from each orientation whose domains match, we get

$$\Phi_B\Phi_C^{-1} = (\Phi_B\Phi_A^{-1}) \circ (\Phi_A\Phi_C^{-1})$$

since the Jacobians of $(\Phi_B\Phi_A^{-1})$ and $(\Phi_A\Phi_C^{-1})$ have negative signs, the Jacobian of their composition must have positive sign, so $\mathcal{B}$ and $\mathcal{C}$ must be compatible, and thus identical.

□

Definition 13.1.13. As with topological manifolds, smooth manifolds with boundary are defined by taking smooth manifolds and replacing the domain of our charts with the upper half-space $\mathbb{H}^n := \{x \in \mathbb{R}^n \mid x_n \geq 0\}$.

13.2 Induced Functional Structures

Example 13.2.1. Consider the torus T^2 defined as the quotient space of $\mathbb{R}^2$ under the equivalence relation identifying points whose coordinates differ by integer amounts (this is fundamentally the same definition as taking the unit square $[0, 1]^2$ with opposite sides identified). Let $q : \mathbb{R}^2 \to T^2$ be the corresponding quotient map.

Finding an atlas on T^2 isnt too complicated: For any open set $U \subseteq \mathbb{R}^2$ with diameter at most one, $q : U \subseteq \mathbb{R}^2 \to q(U) \subseteq T^2$ is a homeomorphism whose inverse we can take as a chart. Since these small diameter sets U still collectively cover $\mathbb{R}^2$, our charts domains $q(U)$ cover T^2.

If $\Phi = (q|_U)^{-1} : q(U) \to U \subseteq \mathbb{R}^n$ and $\Psi = (q|_V)^{-1} : q(V) \to V \subseteq \mathbb{R}^n$ are two charts defined in this way, then we have

$$\Phi\Psi^{-1} : \Psi(q(U) \cap q(V)) \to \Phi(q(U) \cap q(V))$$

Since charts are injective, they respect intersection, giving us

$$\Phi(q(U) \cap q(V)) = \Phi(q(U)) \cap \Phi(q(V)) = U \cap \Phi(q(V))$$
$$\Psi(q(U) \cap q(V)) = \Psi(q(U)) \cap \Psi(q(V)) = \Psi(q(U)) \cap V$$

Thus, Φ^{-1} takes the set $U \cap \Phi(q(V)) \subseteq U \subseteq \mathbb{R}^2$ to its quotient $q(U) \cap q(V)$ in T^2 and then Ψ takes that quotient to the set $\Psi(q(U)) \cap V$. By construction, out original set and our image set must be equivalent, and must thus be translated copies of each other. This tells us that our coordinate change function $\Phi\Psi^{-1}$ are simply translations, and thus smooth.

TODO: Sketch

To find a sensible functional structure on T^2, we can take the following natural definition:

Definition 13.2.2. Let F_X be a functional structure on a topological space X. Let $\Phi : X \to Y$ be continuous. Then the ***induced functional structure*** on Y is given by the finest functional structure such that Φ is a morphism:

$$F_Y(V) = \left\{ f \in C^0(V, \mathbb{R}) \;\middle|\; f \circ \Phi \in F_X(\Phi^{-1}[V]) \right\}$$

Note that this is fundamentally the same idea as the definition of the final topology induced by a continuous function Φ.

We cannot expect this definition to always give a smooth structure on Y - for example, if Y isnt even a *topological* manifold, then any such attempt will obviously fail immediately. For now, we won't concern ourselves with attempts to show when exactly this does end up working out, and instead prove it separately in each application.

In our current example, we want to take the structure induced by q and show that $q|_U : U \subseteq \mathbb{R}^2 \to q[U] \subseteq T^2$ is an isomorphism of functionally structured spaces for sufficiently small open U.

Each $q|_U$ is already a morphism by definition of the induced functional structure, so all we need is a local inverse of $q|_U$.

It suffices to simply take the actual inverse $q|_U^{-1}$, since

$$\begin{aligned} \tau : q^{-1}[q[U]] &\to U \\ x &\mapsto q|_U^{-1}(q(x)) \end{aligned}$$

corresponds to the map collapsing the set $q^{-1}[q[U]]$ of translated copies of U back to the set U itself. Since this map acts as a translation, and thus smoothly, on every connected component of $q^{-1}[q[U]]$, we have $\tau \in C^\infty(q^{-1}[q[U]], U)$.

Thus, for every $f \in C^\infty(U, \mathbb{R})$, we have

$$(f \circ q|_U^{-1}) \circ q = f \circ \tau \in C^\infty(q^{-1}[q[U]], \mathbb{R}),$$

confirming that $f \circ q|_U^{-1} \in F_{T^2}(q[U])$*, and thus that $q|_U^{-1}$ is a morphism.

*Note that $q[U] = (q|_U^{-1})^{-1}[U]$

13.3 Tangent Spaces

Part IV

Algebraic Topology

Chapter 14

Homotopy

Definition 14.0.1

Let X, Y be topological spaces. Let $f, g : X \to Y$ be continuous. Then f is called ***homotopic*** to g via a map h iff. h is a "continuous interpolation between f and g", i.e:

1. h is a continous map $X \times [0,1] \to Y$,
2. For all $x \in X$, we have $h(x,0) = f(x)$,
3. For all $x \in X$, we have $h(x,1) = g(x)$.

We call h a ***free homotopy***, or simply a ***homotopy***, between f and g. We write $f \sim_h g$ if f is a homotopy between f and g. We write $f \sim g$ if there exists a homotopy between f and g.

Exercise 14.0.2. *Show that the property of being homotopic is an equivalence relation.*

Exercise 14.0.3. *Let $f_0, f_1 : X \to Y$ and $g_0, g_1 : Y \to Z$ be continuous. Let h_0 be a homotopy between f_0 and f_1 and let h_1 be a homotopy between f_1 and g_1. Then*

$$H : (x,t) \mapsto h_1(h_0(x,t),t)$$

is a homotopy between $g_0 \circ f_0$ and $g_1 \circ f_1$.

Definition 14.0.4

Let $f : X \to Y$ be continuous. Then f is called a ***homotopy equivalence*** between X and Y iff. there exists a map $g : Y \to X$ such that $g \circ f$ is homotopic to id_X and $f \circ g$ is homotopic to id_Y. We write $X \simeq_{f,g} Y$, $X \simeq_f Y$, or simply $X \simeq Y$. If X and Y are homotopy equivalent, we sometimes say that they have ***the same homotopy type.***

Exercise 14.0.5. *Show that the property of being homotopy equivalent is an equivalence relation.*

Exercise 14.0.6. *Show that homeomorphism implies homotopy equivalence.*

Definition 14.0.7. A topological space X is called ***contractible*** iff. X is homotopy equivalent to a one-point space.

Exercise 14.0.8. 1. *Show that*

$$h : (x, t) \mapsto x - tx$$

is a homotopy between $\mathrm{id}_{\mathbb{R}^n}$ *and* $f : \mathbb{R}^n \mapsto \{0\}$.

2. *Conclude that* $\mathbb{R}^n$ *is contractible.*

3. *Conclude that homotopy equivalence does not imply homeomorphism.*

Exercise 14.0.9. *A topological space* X *is contractible if and only if there exists a point* $x_0 \in X$ *such that* id_X *is homotopic to the constant map* $X \to x_0$.

Definition 14.0.10. Let X be a subset of a topological vector space. Then we call X ***star-shaped*** iff. there exists a point $x_0 \in X$ such that X contains the connecting line between x_0 and any other point $x \in X$, i.e.

$$\begin{aligned} &\forall x \in X : \\ &\forall t \in [0,1] : \\ &(1-t)x_0 + tx \in X \end{aligned}$$

Exercise 14.0.11. *Show that any star-shaped set is contractible. Conclude that any convex set is contractible.*

Lemma 14.0.12. Let X be a topological space. Let $A \subseteq X$. Then A and X are homotopy equivalent if there exists a homotopy between id_X and $\mathrm{id}_X|_A$, i.e. a continuous map $h : X \times [0,1] \to X$ such that for all $x \in X$:

1. $h(x, 0) = x$,
2. $h(x, 1) \in A$,
3. $x \in A \implies h(x, 1) = x$

Definition 14.0.13. Let $A \subseteq X$ be topological spaces. Then A is called a ***deformation retract of X*** if there is a homotopy $h : X \times [0,1] \to X$ such that

1. $h(\cdot, 0) = \mathrm{id}_X$:

$$\forall x \in X : h(x, 0) = x$$

2. $h(\cdot, 1) : X \to A$:

$$\forall x \in X : h(x, 1) \in A$$

3. $h(\cdot, 1)|_A = \mathrm{id}_A$:

$$\forall a \in A : h(a, 1) = a$$

If the last equation can be strengthened to

$$\forall a \in A, t \in [0,1] : h(a, t) = a,$$

we call A a ***strong deformation retract of X***.

Exercise 14.0.14. *Any deformation retract* A *of a topological space* X *is homotopy equivalent to* X.

Exercise 14.0.15. *If $f : X \to Y$ is a continuous function, then the canonical map $r : M_f \to Y$ is a strong deformation retraction.*

14.1 Relative Homotopy

Definition 14.1.1. We say that f and g are ***homotopic relative to a subset $A \subseteq X$*** and write $f \sim_{\mathrm{rel}\, A} g$, $f \sim_{h,\mathrm{rel}\, A} g$ or $f \sim g \,\mathrm{rel}\, A$ iff:

1. $f \sim_h g$,
2. For all $t \in [0,1]$, we have $h(t,\cdot)|_A = f|_A = g|_A$

I personally find the designation "homotopic relative to A" quite unintuitive, given that the property of being homotopic relative to a given subset is stronger than the property of being homotopic.

Exercise 14.1.2. *Show that $f \sim_h g$ is equivalent to $f \sim_{h,\mathrm{rel}\,\emptyset} g$.*

Theorem 14.1.3. Let $\Phi_1, \Phi_2 : [0,1] \to [0,1]$ be homotopies such that $\Phi_1(0) = \Phi_2(0) \in \{0,1\}$ and $\Phi_1(1) = \Phi_2(1) \in \{0,1\}$. Let $F : X \times [0,1] \to Y$ be a homotopy. Let $G_1(x,t) = F(x, \Phi_1(t))$ and $G_2(x,t) = F(x, \Phi_2(t))$. Then we have

$$G_1 \sim_{\mathrm{rel}\, X \times \{0,1\}} G_2$$

Definition 14.1.4. We denote by C a constant homotopy that makes sense in the given context.

Proposition 14.1.5. Given two homotopies $F, G : X \times [0,1] \to Y$ with $F(\cdot,1) = G(\cdot,0)$, their ***concatenation***

$$(F * G)(x,t) = \begin{cases} F(x,2t) & 0 \le t \le 1/2, \\ G(x,2t-1) & 1/2 \le t \le 1. \end{cases}$$

is again a homotopy $F * G : X \times [0,1] \to Y$.

Note that the order here doesn't match the order in function composition - we go through F first, then through G.

Theorem 14.1.6. We have $C * F \sim F * C \sim F \,\mathrm{rel}\, X \times \{0,1\}$

Definition 14.1.7. Let $h : X \times [0, 1 \to Y]$ be a homotopy. We define the ***inverse of h*** to be the homotopy that "moves through h backwards", i.e.

$$h^{-1}(x,t) = \overline{h}(x,t) := h(x, 1-t)$$

Theorem 14.1.8. Given a homotopy h, we have

$$h * h^{-1} \sim C \,\mathrm{rel}\, X \times \{0,1\},$$

where C is the constant homotopy $C(x,t) = h(x,0)$

14.2 The Fundamental Group

Definition 14.2.1. Let X be a topological space. Then a continuous map $\gamma[0,1] \to X$ with $\gamma(0) = \gamma(1)$ is known as a ***loop***. We denote the set of all loops in X with $\gamma(0) = \gamma(1) = x_0$ as $\Omega(X, x_0)$.

Lemma 14.2.2. Let $\gamma : [0,1] \to X$ be a loop. Let $\varphi : [0,1] \to [0,1]$ be a continuous map such that $\gamma(0) = 0$ and $\gamma(1) = 1$. Then $\gamma \circ \varphi$ and γ are homotopic relative to $\{0,1\}$.

Proof. The map

$$h : (x,t) \mapsto \gamma((1-t)x + t\varphi(x))$$

is a homotopy between γ and $\gamma \circ \varphi$. □

Definition 14.2.3. Let $f, g \in \Omega(X, x_0)$. Then we define the ***concatenation of f and g*** as the map

$$(f * g)(t) = \begin{cases} f(2t) & 0 \le t \le 1/2, \\ g(2t-1) & 1/2 \le t \le 1. \end{cases}$$

Definition 14.2.4. Let $\gamma \in \Omega(X, x_0)$. We define the ***inverse loop*** of γ to be the loop that "moves through gamma backwards", i.e.

$$\overline{\gamma}(t) := \gamma(1-t)$$

Exercise 14.2.5. *Verify that* $\overline{\overline{\gamma}} = \gamma$.

Exercise 14.2.6. *Let* $\gamma \in \Omega(X, x_0)$. *Then* $\gamma * \overline{\gamma}$ *is homotopic to* $e_{x_0} : X \to \{x_0\}$ *relative to* $\{0,1\}$.

Theorem 14.2.7: Existence of the fundamental group

Let X be a topological space. Let $x_0 \in X$. Then the quotient space

$$\pi_1(X, x_0) = \Omega(X, x_0)/\sim_{\mathrm{rel}(0,1)}$$

of loops modulo homotopy becomes a group, known as the ***fundamental group of*** (X, x_0), when equipped path concatenation

$$[f] * [g] = [f * g],$$

as the group operation. The neutral element of this group is $e_{x_0} := (x \mapsto x_0)$.

Theorem 14.2.8. If we additionally define

$$\pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$$
$$[\gamma] \mapsto [f \circ \gamma]$$

for all continuous $f : X \to Y$, the map π_1 becomes a covariant functor from the category of topological spaces into the category of groups.

Definition 14.2.9. Whenever we need to make the base point clear, we write $\pi_1(f, x) := \pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$

Theorem 14.2.10: Fundamental groups at different base points

Let X be a topological space. Let $x, y \in X$. Then every continuous path $\delta : [0, 1] \to X$ induces a group isomorphism

$$J_\delta : \pi_1(X, \delta(0)) \cong \pi_1(X, \delta(1))$$
$$[\gamma] \mapsto \left[\overline{\delta} * \gamma * \delta\right]$$

Proof.

1. Since loop concatenation preserves homotopy, J_δ sends different representatives of the same equivalence class to the same equivalence class.

2. J_δ goes from $\pi_1(X, x)$ to $\pi_1(X, y)$, since for any $\gamma : [0, 1] \to X$ with $\gamma(0) = \gamma(1) = x$, $\overline{\delta}$ traces out a path from $\delta(1)$ to $\delta(0)$, then γ traces out a loop from $\delta(0)$ to $\delta(0)$, and then δ traces out a path from $\delta(0)$ back to $\delta(1)$, i.e. $\overline{\delta} * \gamma * \delta$ is a loop from $\delta(1)$ to $\delta(1)$.

3. J_δ is a group homomorphism, since

$$\begin{aligned} J_\delta([\gamma_1 * \gamma_2]) &= \left[\overline{\delta} * \gamma_1 * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * e_{x_0} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta * \overline{\delta} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta\right] * \left[\overline{\delta} * \gamma_2 * \delta\right] \\ &= J_\delta([\gamma_1]) * J_\delta([\gamma_2]) \end{aligned}$$

   (Recall that any map that respects the group operation also respects the neutral element and inverses, and is thus a homomorphism.)

4. J_δ is bijective with $J_\delta^{-1} = J_{\overline{\delta}}$:

$$J_\delta(J_{\overline{\delta}}(\gamma)) = \left[\overline{\delta} * \left(\delta * \gamma * \overline{\delta}\right) * \delta\right] = [\gamma]$$
$$J_{\overline{\delta}}(J_\delta(\gamma)) = \left[\delta * \left(\overline{\delta} * \gamma * \delta\right) * \overline{\delta}\right] = [\gamma]$$

□

Corollary 14.2.11. If X is path connected, $\pi_1(X, x) \cong \pi_1(X, y)$ for any $x, y \in X$.

Definition 14.2.12. Let $\gamma_0 \in \Omega(X, x)$ an $\gamma_1 \in \Omega(X, y)$. Then we call $h : [0, 1] \times [0, 1] \to X$ a ***free homotopy of loops***, or simply ***loop homotopy***, between γ_0 and γ_1 iff. h is a homotopy between γ_0 and γ_1 such that $h_t(x) := h(x, t)$ is a loop for any fixed $t \in [0, 1]$.

Exercise 14.2.13. *1. Verify that h is a loop homotopy iff. h is a homotopy and $h_t(0) = h_t(1)$ for any $t \in [0, 1]$.*

2. *Show that any $\{0,1\}$-homotopy is a loop homotopy.*

The additional constraint that and $h(\cdot, t)$ is a loop may seem superfluous, but there are in fact homotopies between loops that aren't loop homotopies:

Exercise 14.2.14. 1. *Let $\gamma_0 = \gamma_1 : [0,1] \to \{0\}$ be constant loops. Let $h(x,t) := xt(1-t)$. Then h is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

2. *Let $\gamma_0(x) = e^{2\pi i x}$. Let $\gamma_1(x) = e^{4\pi i x}$. Then $h(x,t) := e^{2\pi i(1+t)x}$ is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

3. *Assume that any loop $\gamma : [0,1] \to S^1$ with 1 as its base point is homotopic a loop of the form $\gamma(x) := e^{2\pi n i x}$ for some $n \in \mathbb{Z}$ (we will prove this later). Does 2. imply that the fundamental group $\pi_1(S^1, 1)$ is trivial?*

Definition 14.2.15. A loop $\gamma \in \Omega(X,x)$ is called ***contractible in X*** iff. there exists an $x \in X$ and a loop homotopy h such that $\gamma \sim_h e_x$.

Lemma 14.2.16. $\gamma \in \Omega(X,x)$ is contractible in X if $\gamma \sim_{\mathrm{rel}\,0,1} e_{\gamma(0)}$.

Theorem 14.2.17. Let $\gamma \in \Omega(X,x)$ be a loop. Let $S^1 \cong I/\{0,1\}$ be equipped with the quotient topology. Then γ is contractible if and only if $\gamma \circ p : S^1 \to X$ has a continuous extension to the closed unit ball $\overline{B}^2$.

Theorem 14.2.18. Let h be a homotopy between $f, g : X \to Y$. Let $x_1 \in X$. Then

$$\begin{aligned} h(x_1, \cdot) : [0,1] &\to Y \\ t &\mapsto h(x_1, t) \end{aligned}$$

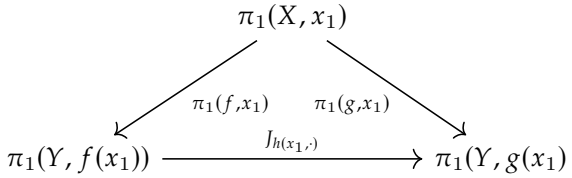
is a path from $f(x_1)$ to $g(x_1)$ which fulfills

$$\pi_1(g, x_1) = J_{h(x_1,\cdot)} \circ \pi_1(f, x_1)$$

Recall that $\pi_1(g, x_1)$ is the map $[\gamma] \mapsto [g \circ \gamma]$, and that J_δ is the map $[\gamma] \mapsto [\overline{\delta} * \gamma * \delta]$, so this theorem states that

$$[g \circ \gamma] = \left[\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)\right]$$

for any $\gamma \in \pi_1(X, x_1)$, or, equivalently, that the diagram



commutes.

Proof. Define

$$H_t(x) := h(x_1, t + (1-t)x).$$

We have $(f \circ \gamma) \in \Omega(X, f(x_1)$ and $(g \circ \gamma) \in \Omega(X, g(x_1))$, since

$$\begin{aligned} f(\gamma(0)) &= f(x_1) = f(\gamma(1)) \\ g(\gamma(0)) &= g(x_1) = g(\gamma(1)) \end{aligned}$$

Then a homotopy between $g \circ \gamma$ and $\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)$ is given by:

$$k(x,t) := (\overline{H_t} * (h(\cdot, t) \circ \gamma) * H_t)(x),$$

since we get:

$$\begin{aligned} k(x,0) &= (\overline{H_0} * (h(\cdot, 0) \circ \gamma) * H_0)(x) \\ &= (\overline{H_0} * (f \circ \gamma) * H_0)(x) \\ &= (\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot))(x) \end{aligned}$$

and

$$\begin{aligned} k(x,1) &= (\overline{H_1} * (h(\cdot, 1) \circ \gamma) * H_1)(x) \\ &= (h(x_1, (1-1) + (1-(1-1))(\cdot)) * (g \circ \gamma) * h(x_1, 1 + (1-1)(\cdot)))(x) \\ &= (h(x_1, \cdot) * (g \circ \gamma) * h(x_1, 1))(x) \\ &= (e_{g(x_1)} * (g \circ \gamma) * e_{g(x_1)})(x) \\ &= (g \circ \gamma)(x), \end{aligned}$$

where the second to last step follows since the loop $(g \circ \gamma)$ ends on the base point $g(x_1)$. □

Lemma 14.2.19. Let $\gamma : [0,1] \to X$ be continuous. Let $f : X \to Y$. Then

$$\pi_1(f, \gamma(1)) \circ J_\gamma = J_{f \circ \gamma} \circ \pi_1(f, \gamma(0)),$$

i.e. the diagram

$$\begin{array}{ccc} \pi_1(X, \gamma(0)) & \xrightarrow{\pi_1(f, \gamma(0))} & \pi_1(Y, f(\gamma(0))) \\ \downarrow{\scriptstyle J_\gamma} & & \downarrow{\scriptstyle J_{f \circ \gamma}} \\ \pi_1(X, \gamma(1)) & \xrightarrow{\pi_1(f, \gamma(1))} & \pi_1(Y, f(\gamma(1))) \end{array}$$

commutes.

Proof. Let $\sigma \in \Omega(X, \gamma(0))$. Then we have

$$\begin{aligned} \pi_1(f, \gamma(1)) \circ J_\gamma([\sigma]) &= \pi_1(f, \gamma(1))([\overline{\gamma} * \sigma * \gamma]) \\ &= \left[f \circ (\overline{\gamma} * \sigma * \gamma)\right] \\ &= \left[(f \circ \overline{\gamma}) * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= \left[\overline{(f \circ \gamma)} * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= J_{f \circ \gamma}([f \circ \sigma]) \\ &= J_{f \circ \gamma} \circ \pi_1(f, \gamma(0))([\sigma]) \end{aligned}$$

□

Theorem 14.2.20. Let $f : X \to Y$ be a homotopy equivalence. Let $x_0 \in X$. Then the map $\pi_1(f, x_0)$ is a group isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.

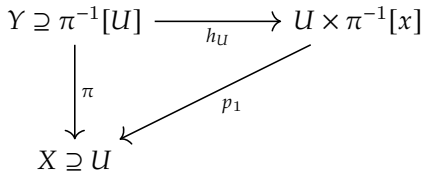
Chapter 15

Fibrations and Coverings

Definition 15.0.1: Locally Trivial Fibration

Let $\pi : Y \to X$ be continuous. Then π is called a ***locally trivial fibration*** iff. for every $x \in X$, there exists an open neighborhood U such that there exists a homeomorphism $h_u : \pi^{-1}[U] \cong U \times \pi^{-1}[x]$ fulfilling $p_1 \circ h_U = \pi|_{\pi^{-1}[U]}$.

$\pi^{-1}[x]$ is called the ***Fiber of π at x***, and sometimes written F_x.



Definition 15.0.2. Let $\pi : Y \to X$ be a locally trivial fibration. Then:

1. If all $\pi^{-1}[x]$ are homeomorphic, we call π a ***fiber bundle***.
2. If every $\pi^{-1}[x]$ is discrete, we call π a ***covering map***.
3. If we can pick $U = X$, π is called a ***trivial fibration***.

Exercise 15.0.3. *If we replace $\pi^{-1}[x]$ in the definition by an arbitrary topological space F_U, we get $F_U \cong \pi^{-1}[x]$. Thus, both definitions are equivalent.*

Lemma 15.0.4. Let $\pi : Y \to X$ be a locally trivial fibration. Let X be connected. Then π is a trivial fibration.

Proof. Let $x_0 \in X$. Let $V = \{x \in X \mid F_x \cong F_{x_0}\}$ be the set of all points whose fibers are homeomorphic to the fiber of x_0. Let U be a neighborhood as in the definition. Then the fibers of all $x \in U$ must be homeomorphic by definition. Thus, every point in V has a neighborhood contained in V, so V is open. We have

$$X \setminus V = \bigcup_{y \notin V} \left\{x \in X \,\middle|\, F_x \cong F_y\right\} := \bigcup_{y \notin V} V_y$$

Since every V_y is open by the same argument, $X \setminus V$ must be open. But since X is connected with $(X \setminus V) \cup V = X$ and V nonempty, $X \setminus V$ must

be empty. □

Theorem 15.0.5. A continuous function $\pi : Y \to X$ is a covering map if and only if every $x \in X$ has an open neighborhood U such that $\pi^{-1}[U]$ can be decomposed into disjoint open sets $V_i \subseteq Y$ such that $\pi|_{V_i}$ is a homeomorphism $V_i \cong U$.

Proof. $\Rightarrow$: Let π be a covering map. Let $I = F_U$. Since h_U is bijective, the sets $V_i := h_U^{-1}[U \times \{i\}]$ are disjoint. Since h_U is a homeomorphism and $I = F_U$ is discrete, the V_i are also open. Since $p_1|_{U\times\{i\}}$ is a homeomorphism $U \times \{i\} \cong U$, the composition

$$\pi|_{V_i} = (p_1|_{U\times\{i\}}) \circ h|_{V_i}$$

must be a homeomorphism $V_i \mapsto U$.

$\Leftarrow$: Define

$$\begin{aligned} h_U : \pi^{-1}[U] &\cong U \times I \\ y \in V_i &\mapsto (\pi(y), i). \end{aligned}$$

Then the restriction of h_U to each V_i is by definition a homeomorphism $V_i \cong U \cong U \times \{i\}$, and so the whole of h_U is a homeomorphism $\pi^{-1}[U] = \bigsqcup_{i\in I} V_i \cong \bigsqcup_{i\in I} U \times \{i\} = U \times I$.

□

Definition 15.0.6: Lifting

Let $\pi : Y \to X$ be a locally trivial fibration. Let $f : Z \to X$ be continuous. Then a continuous map $\hat{f} : Z \to Y$ is called a ***lift from f to Y*** if $\pi \circ \hat{f} = f$.

Any path can be lifted:

Theorem 15.0.7. Let $\pi : Y \to X$ be a locally trivial fibration. Let $\gamma : [0, 1] \to XX$ be continuous. Let $\pi(y_0) = \gamma(0)$. Then there exists a lift $\hat{\gamma} : [0, y] \to Y$ with $\hat{\gamma}(0) = y_0$ such that $\pi \circ \hat{f} = f$.

And lifts respect fundamental groups:

Theorem 15.0.8. Let $\pi : Y \to X$ be a locally trivial fibrations. Let $f : Z \to X$. Let $\hat{f} : Z \to Y$ be a lift of f. Then we have:

$$\begin{aligned} &\forall z_0 \in Z : \\ &\exists y_0 \in Y : \\ &\quad \hat{f}(z_0) = y_0, \\ &\quad \pi_1(f)[\pi_1(Z, z_0)] \subseteq \pi_1(\pi)[\pi_1(Y, y_0)] \end{aligned}$$

Proof.

$$\begin{aligned}\pi_1(f)[\pi_1(Z, z_0)] &= (\pi_1(\pi \circ \hat{f}))[\pi_1(Z, z_0)] \\ &= (\pi_1(\pi) \circ \pi_1(\hat{f}))[\pi_1(Z, z_0)] \\ &= \end{aligned}$$

□